ONTARIO TEJ4M COMPUTER ENGINEERING TECHNOLOGY

PRELIMINARY DESIGN DOCUMENT

PROJECT

Bland2Grand

Smart Autonomous Carousel Spice Dispensing Appliance

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Version	1.0
Date	March 2026

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1 Introduction

1.1 Project Overview

Bland2Grand is a smart, autonomous countertop spice dispensing appliance designed to remove the guesswork from seasoning food. A user types the name of any dish into a mobile web application, the system retrieves matching spice blend options from a local recipe database, the user selects a blend and a serving count, and the machine autonomously dispenses the precise gram-accurate quantity of each required spice directly into a bowl placed below — with no manual measurement required at any step.

The physical machine is centred on a motorised octagonal carousel. Eight interchangeable trapezoidal spice containers are seated one on each face of the octagon. The carousel is driven by a NEMA 23 stepper motor (M1) mounted centrally below the platform. M1 turns a small pinion gear on its shaft, which meshes externally with a larger ring gear that forms the outer circumference of the carousel platform. Because the gears are in external mesh, the carousel rotates in the opposite direction to the motor shaft. An AS5600 12-bit absolute magnetic encoder (U6) mounted coaxially on the M1 shaft provides closed-loop angular position feedback, allowing the firmware to verify that each of the eight 45° index positions has been correctly reached.

Once the correct container is indexed to the active position, a NEMA 17 stepper motor (M2) fixed to the tower frame directly below the carousel plane, with its shaft pointing upward, drives the dispensing mechanism. M2's shaft carries a *half spur gear* — a spur gear with teeth machined on only 180° of its circumference. Each spice container has a full spur gear exposed on its bottom face, connected internally to a horizontal hex shaft that runs through the container body and turns the auger screw inside. As the carousel brings a container to the active position, the container's full spur gear contacts M2's half spur gear. When M2 rotates, the toothed half engages the container's spur gear and transfers rotation through 90° to the hex shaft, turning the auger, which pushes spice through the container's downward-facing exit into the bowl below. When M2 stops, the toothless half faces the container's spur gear, providing a positive mechanical cutoff: no torque is transmitted and no passive dripping can occur.

The bowl sits on top of a NOYITO 1 kg load cell (LC1) paired with an HX711 24-bit ADC amplifier module (U3) to provide real-time gram-accurate weight feedback. An Arduino Uno R4 WiFi (U1) coordinates all motion, sensing, and wireless communication with a Flask web server running on a host laptop over the onboard 2.4 GHz 802.11 b/g/n WiFi module.

1.2 Document Purpose and Scope

A complete Hardware Requirements Specification (HRS) for this project was written separately. The HRS defines all normative *shall* requirements and serves as the authoritative design contract between the student designer (Elliott Starosta) and the course teacher (Mr. Roller). This Preliminary Design Document (PDD) describes *how* those requirements will be met. Specifically, it provides:

- A high-level system block diagram, more detailed than the HRS version and intended as the authoritative reference for evaluation;
- Detailed sub-circuit descriptions supported by engineering theory and calculations;
- Internal and external interface signal tables;
- Pseudocode for both the Arduino C++ firmware and the Python Flask backend;
- The complete project schedule and Gantt chart;
- A full parts list with sourcing links.

The HRS is available alongside this document for reference.

Design Revision Note: Logic Level Shifting

During the initial drafting of the HRS, a logic level shifter between U1 and the TB6600 drivers (U4, U5) was included as a deliberate conservative measure: without yet knowing the exact TB6600 variant that would be sourced, and with the generic TB6600 datasheet citing a 3.5 V high-level input threshold, erring on the side of caution was the correct approach at the requirements stage.
During PDD preparation, the design was revisited with the actual sourced components in hand. The Funsto TB6600 units used for U4 and U5 specify a control signal input range of 3.3–24 V, which directly accommodates the RA4M1 output. Separately, confirming that the Uno R4 WiFi supplies 5 V to the RA4M1 V_{CC} pin places $V_{OH,\min}$ at 4.2 V per RA4M1 datasheet Table 2.7 — providing further margin. With specific component specifications now confirmed, the level shifter is no longer required and has been removed from the design. **HRS-3.3.1.1, HRS-3.3.1.2, and HRS-3.3.1.7 are superseded by this finding.**

2 Product Design

2.1 System Block Diagram

Figure 1 shows the complete Bland2Grand system partitioned into its major functional sub-circuits. Power flows downward from the 24 V supply (BT1) through the distribution bus to the motor drivers and buck converter. Control signals flow horizontally between the Arduino (U1) and each peripheral. The weight feedback path (LC1 $\rightarrow$ U3 $\rightarrow$ U1) closes the gram-accuracy loop, while WiFi bridges U1 to the Flask server and the user's phone.

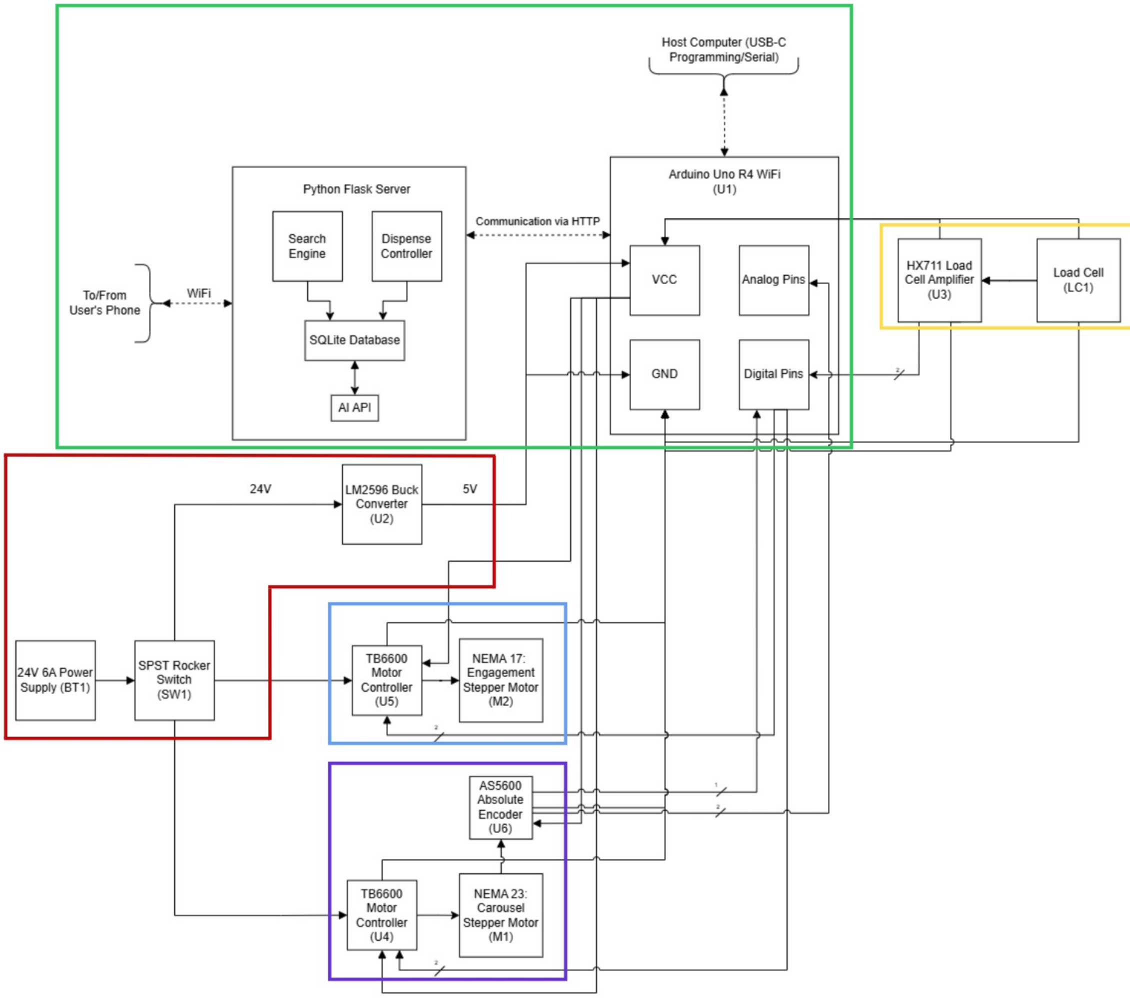


Figure 1: Bland2Grand high-level system block diagram, partitioned into its five major sub-circuits. Power distribution is shown in red; motor control in blue; sensor feedback in green; software/communication in orange.

2.2 Sub-Circuit Descriptions

Each sub-circuit below corresponds to a labelled functional block in Figure 1. They are presented in the order that power and signals flow through the system:

1. **Power System** (Sub-Circuit 1) — mains AC enters and establishes the 24 V, 5 V, and 3.3 V rails that every other block depends on.
2. **Carousel Drive** (Sub-Circuit 2) — uses the 24 V rail to rotate the octagonal platform and position the correct spice container at the dispensing point.
3. **Auger Engagement Drive** (Sub-Circuit 3) — uses the same 24 V rail to meter spice from the indexed container.
4. **Weight Sensing** (Sub-Circuit 4) — closes the gram-accuracy loop by feeding real-time

mass data back to the Arduino over the 5 V logic rail.

5. **Software and Communication** (Sub-Circuit 5) — ties all hardware together through firmware running on U1 and the Flask server communicating over WiFi.

2.2.1 Sub-Circuit 1: Power System (BT1, U2)

The power system is the foundation of the entire Bland2Grand design. Every other sub-circuit depends on the voltage rails established here, so the power architecture was chosen to be simple, robust, and immediately shut-down-able from a single switch.

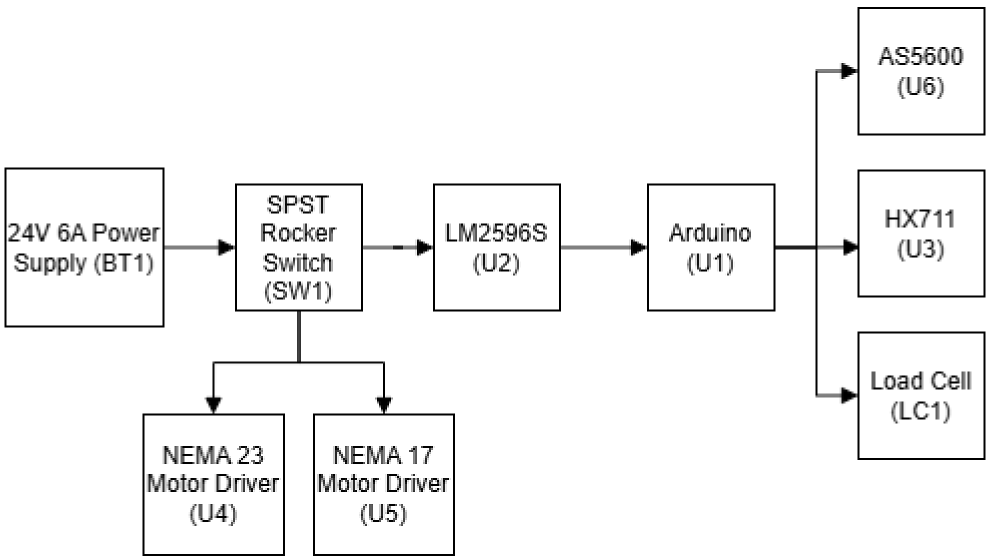


Figure 2: Power System block extracted from the high-level block diagram. A single 24 V 6 A supply (BT1) feeds the system through a series-connected SPST rocker switch (SW1), which controls power to the entire 24 V distribution bus. This bus supplies both TB6600 stepper drivers directly. The LM2596S-ADJ buck converter (U2) steps 24 V down to a regulated 5 V logic rail for U1 and U3, while U6 draws 3.3 V from U1's onboard regulator.

Architecture Overview. The power architecture of Bland2Grand is centred around a single 24 V, 6 A AC/DC switching supply (BT1) and a single-pole single-throw (SPST) rocker switch (SW1) mounted on the tower enclosure. SW1 is placed in series with the positive 24 V line, between BT1's barrel jack output and the system's internal 24 V distribution bus.

When SW1 is opened, power is removed from the entire system simultaneously. Both TB6600 stepper motor drivers (U4 and U5) lose their motor supply voltage; U2 loses its input and consequently shuts down the 5 V rail that powers U1 and U3; and U6 loses its 3.3 V supply derived from U1. This design enables an immediate and complete hardware-level shutdown, eliminating the need for any software-controlled power-down sequence.

SW1 is rated for 20 A at 24 V DC, providing a fivefold safety margin over the system's worst-case current draw of 3.7 A. The use of a 24 V supply was chosen deliberately: Both

TB6600 drivers operate within a 9–42 V range, and running at 24 V maximises available torque without requiring an additional high-voltage rail. BT1 features a universal 100–240 V AC input and a centre-positive 5.5×2.1 mm barrel jack output.

Wiring and Voltage Drop Analysis. All 24 V distribution wiring uses 18 AWG stranded wire. The worst-case power run is 300 mm at 2.0 A to driver U4. Using the wire resistance of 0.02093 Ω/m:

$$V_{\text{drop}} = I \times R = 2.0 \times (0.02093 \times 0.300) = 12.6\,\text{mV}$$

A 12.6 mV drop against a 24 V rail is 0.052% — entirely negligible. Signal wiring uses 22 AWG solid-core jumper wire throughout; currents on signal lines are in the microampere range and produce no measurable voltage drop at these gauges.

Buck Converter (U2): 24 V to 5 V. The Arduino Uno R4 WiFi (U1) and HX711 ADC module (U3) both require a regulated 5 V supply. Deriving this directly from 24 V with a linear regulator would dissipate $(24 - 5) \times 0.5 = 9.5$ W as heat and require a substantial heatsink. Instead, the MECCANIXITY LM2596S-ADJ DC-DC buck converter module (U2) steps 24 V down to 5 V with a typical efficiency of 85–90%, dissipating only about 0.3 W.

The LM2596S is a 150 kHz switching step-down regulator. Internally, an error amplifier maintains a fixed reference voltage of $V_{\text{ref}} = 1.23$ V on the ADJ feedback pin. An external resistor divider from the output rail to GND, with the ADJ pin at the midpoint, sets the desired output:

$$V_{\text{out}} = V_{\text{ref}} \times \left(1 + \frac{R_2}{R_1}\right) = 1.23 \times \left(1 + \frac{R_2}{R_1}\right)$$

Solving for $V_{\text{out}} = 5.0$ V:

$$\frac{R_2}{R_1} = \frac{V_{\text{out}}}{V_{\text{ref}}} - 1 = \frac{5.0}{1.23} - 1 = 3.065$$

On the MECCANIXITY module, this ratio is set by rotating an onboard trimmer potentiometer. The output is verified with a multimeter and adjusted to exactly 5.0 V before any load is connected. All required passives (the Schottky catch diode, 68 μH output inductor, 100 μF input capacitor, and 220 μF output capacitor) are integrated onto the module; no external components are needed.

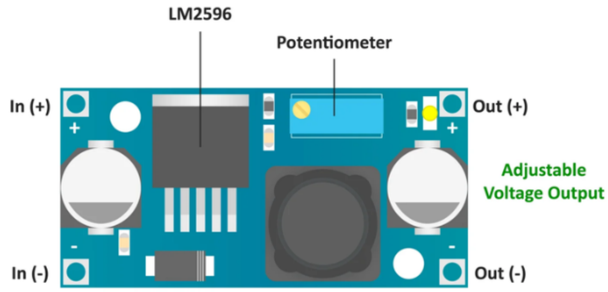


Figure 3: U2 MECCANIXITY LM2596S-ADJ module. The onboard trimmer potentiometer sets the output to exactly 5.0 V. All required passives (Schottky diode, inductor, filter capacitors) are integrated; no external components are needed.

3.3 V Rail. The AS5600 encoder (U6) requires 3.3 V and draws only 10 mA. Rather than adding a third regulated rail, it is sourced from U1's onboard 3.3 V linear regulator, which can supply up to 50 mA and is itself fed from the 5 V U2 output. This keeps the 3.3 V domain entirely within U1 and simplifies the wiring harness.

Worst-Case Power Budget. The table below summarises the full system power budget under worst-case simultaneous loading.

Component	Rail	Current (max)	Powered By
M1 (NEMA 23) via U4 (set 2.0 A)	24 V	2000 mA	24 V rail
M2 (NEMA 17) via U5 (set 1.5 A)	24 V	1500 mA	24 V rail
U2 quiescent + switching loss	24 V	≈200 mA	24 V rail
U1 (Uno R4 WiFi, MCU + WiFi active)	5 V	500 mA	U2 output
U3 (HX711 ADC module)	5 V	2 mA	U2 output
U6 (AS5600 encoder, via U1 3.3 V reg)	3.3 V	10 mA	U1 onboard reg
Total 24 V draw (worst case)	**24 V**	**3700 mA**	**BT1 (6 A rated)**
Headroom	—	**2300 mA**	**38% reserve**

The 6 A supply operates at 62% of rated capacity in the worst case. The 5 V load on U2 is $I_{5V} = 500 + 2 = 502$ mA against its 3 A output rating — 16.7% utilisation, providing 83.3% thermal headroom with no heatsink required on the module. The 3.3 V encoder load of 10 mA consumes 20% of U1's 50 mA regulator budget, leaving 40 mA available for any

future 3.3 V peripherals. Every component operates well within its rated thermal envelope from this single supply.

2.2.2 Sub-Circuit 2: Carousel Drive (M1, U4, U6)

The carousel drive sub-circuit is responsible for rotating the octagonal spice container platform to index any of the eight containers to the fixed dispensing position. The signal path begins at U1, passes through the TB6600 microstepping driver (U4) to the NEMA 23 stepper motor (M1), through a pinion-and-ring gear train to the carousel platform, and closes the loop back to U1 via the AS5600 absolute magnetic encoder (U6) reading the M1 shaft.

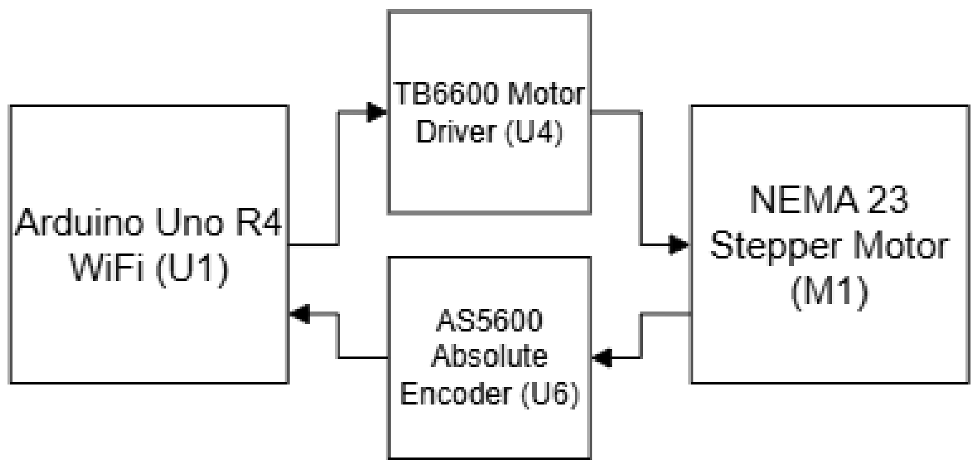


Figure 4: Carousel Drive block extracted from the high-level block diagram. U1 digital pins D3 and D4 send STEP and DIR pulses to TB6600 driver U4 over the 24 V rail. U4 drives the NEMA 23 motor M1, which turns the carousel through the pinion-and-ring gear train. AS5600 encoder U6 reads the M1 shaft angle over I2C and reports it back to U1 on pins A4 and A5.

Stepper Motor Theory and Motor Selection. A stepper motor is a brushless DC motor whose rotor is divided into a fixed number of equally spaced magnetic poles. When the two coil phases (A and B) are energised in sequence, the rotor steps to align with each successive stator pole. The 23HS5628 NEMA 23 has a step angle of 1.8°, producing exactly 200 full steps per mechanical revolution. Unlike a DC motor, a stepper holds its position between steps without an encoder, providing passive holding torque to keep the loaded carousel stationary between index moves.

Parameter	Value
Step angle	1.8° (200 full steps/rev)
Rated phase current	2.8 A
Holding torque (at rated current)	126 N·cm
Output shaft diameter	8 mm
Phase resistance / inductance	0.9 Ω / 2.5 mH

NEMA 23 was chosen over NEMA 17 because the carousel platform carries the mass of eight loaded spice containers. Estimating each loaded container at 125 g (PLA shell plus spice fill), the total rotating load is $m = 8 \times 0.125 = 1.0$ kg. Treating the containers as point masses at a representative radius of $r = 80$ mm from the shaft centre, the rotational inertia is:

$$J_{\text{load}} = mr^2 = 1.0 \times (0.080)^2 = 6.4 \times 10^{-3}\ \text{kg m}^2$$

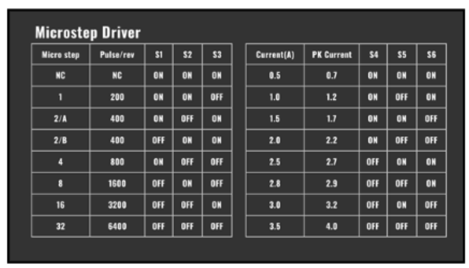
A NEMA 17 (17HS4401S) provides only 45 N·cm of holding torque. The NEMA 23 provides 126 N·cm — a 2.8$\times$ margin that ensures reliable indexing under full load without step loss. The TB6600 current limit for U4 is set to 2.0 A rather than the motor's full 2.8 A rating; this is sufficient for the light positioning load while reducing steady-state thermal dissipation in the driver. Because the carousel must hold a loaded platform stationary between index moves, the ENA pin of U4 is tied permanently low (always enabled), keeping both coils energised and full holding torque applied at all times.

TB6600 Driver (U4) and Microstepping. The TB6600 (U4) accepts a step/direction interface from U1 and converts it into the bipolar phase current waveforms required by M1's two coil pairs. Each rising edge on the PUL$-$ input advances M1 by one microstep in the direction set by the DIR$-$ input. U1 drives both signals active-low using common-anode wiring: The PUL$+$, DIR$+$, and ENA$+$ terminals are all tied to the +5 V rail as a pull-up reference, and U1 pulls the corresponding $-$ terminal LOW to assert each signal. The Funsto TB6600 units sourced for this project specify a control signal input range of 3.3–24 V, which directly accommodates the RA4M1's 3.3 V output without any level shifting required.

U4 Pin	Connection
ENA+	+5 V
ENA−	GND (always enabled — continuous holding torque)
PUL+	+5 V
PUL−	U1 pin D3
DIR+	+5 V
DIR−	U1 pin D4
VCC	+24 V rail
GND	System GND
A+, A−	M1 coil A
B+, B−	M1 coil B

The DIP switches on U4 are set to 1/8 microstepping, dividing each full 1.8° step into 8 equal microsteps by shaping the coil current as a sinusoidal staircase rather than a square wave. This reduces mechanical resonance and produces smoother carousel motion. The effective motor shaft resolution becomes:

$$\text{Microsteps/rev} = 200 \times 8 = 1600 \text{ microsteps/rev at motor shaft}$$


Microstep Driver

Micro step	Pulse/rev	S1	S2	S3
NC	NC	ON	ON	ON
1	200	ON	ON	OFF
2/A	400	ON	OFF	ON
2/B	400	OFF	ON	ON
4	800	ON	OFF	OFF
8	1600	OFF	ON	OFF
16	3200	OFF	OFF	ON
32	6400	OFF	OFF	OFF

Current(A)	PK Current	S4	S5	S6
0.5	0.7	ON	ON	ON
1.0	1.2	ON	OFF	ON
1.5	1.7	ON	ON	OFF
2.0	2.2	ON	OFF	OFF
2.5	2.7	OFF	ON	ON
2.8	2.9	OFF	OFF	ON
3.0	3.2	OFF	ON	OFF
3.5	4.0	OFF	OFF	OFF

Figure 5: TB6600 driver U4 DIP switch panel configured to 2.0 A current limit and 1/8 microstepping. The same switch layout is used on U5, with the current limit set to 1.5 A.

Gear Train and Carousel Kinematics. M1's 8 mm output shaft carries a small pinion gear that meshes externally with a larger ring gear forming the outer rim of the carousel platform. In an external gear mesh the output always rotates in the opposite direction to the input, so the carousel turns counter-clockwise when M1 turns clockwise. The gear ratio is:

$$\text{GR} = \frac{N_{\text{ring}}}{N_{\text{pinion}}} = \frac{96}{48} = 2$$

Every full M1 revolution therefore advances the carousel by $360/2 = 180°$. The eight container positions are spaced at 45° carousel intervals; each 45° index move requires

$45 \times 2 = 90°$ of M1 shaft rotation. At 1600 microsteps/rev, this corresponds to:

$$N_{\text{steps,index}} = \frac{90}{360} \times 1600 = 400 \text{ microsteps per index move}$$

At a motor shaft speed of 60 RPM, one index move takes:

$$t_{\text{index}} = \frac{90}{360} \times \frac{60\,\text{s/min}}{60\,\text{RPM}} = \frac{1}{4}\,\text{s} = 0.25\,\text{s per index move}$$

A full 8-spice blend requires at most 7 index moves. In the worst case, total carousel travel time is $7 \times 0.25 = 1.75$ s — a negligible fraction of the 120 s total dispense budget.

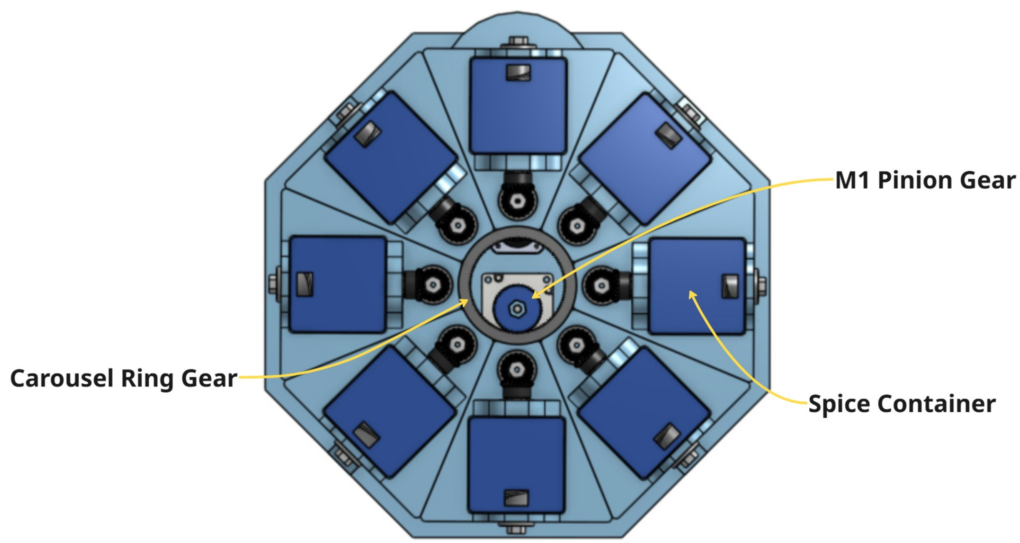


Figure 6: Top-view CAD render of the carousel platform, showing the ring gear on the outer rim, the eight trapezoidal container bays, and the central bore for the M1 shaft and pinion.

AS5600 Encoder (U6) and Closed-Loop Position Verification. Open-loop step counting alone is insufficient for reliable positioning: mechanical resonance, inertia overshoot, or a momentary load spike can cause the motor to miss steps without the controller knowing. The AS5600 (U6) resolves this by providing absolute angular feedback of the M1 shaft. It is mounted coaxially on the M1 shaft end with a diametrically magnetised disc magnet press-fit to the shaft within a 0.5 mm axial air gap. The AS5600 communicates over I2C at a fixed address of 0x36, connected to U1 pins A4 (SDA) and A5 (SCL) at 400 kHz. The module has onboard 4.7 kΩ pull-up resistors on both bus lines; no external pull-ups are required.

The AS5600 outputs a raw 12-bit angle value from 0 to 4095 covering one full shaft revolution:

$$\text{Resolution} = \frac{360}{2^{12}} = \frac{360}{4096} = 0.088 \text{ per count}$$

Because U6 reads the M1 shaft, a 45° carousel index corresponds to $45 \times 2 = 90°$ of M1 shaft rotation, mapping to:

$$\Delta\theta_{\text{counts}} = \frac{90}{360} \times 4096 = 1024 \text{ counts per index move}$$

The firmware position tolerance is $\pm 1.0°$ at the carousel, equal to $\pm 2°$ at the M1 shaft:

$$\text{Tolerance} = \pm\frac{2}{360} \times 4096 \approx 23 \text{ counts}$$

A ± 23 count tolerance against a 1024-count index step is a noise margin better than 2.2% — more than adequate to detect any real positioning error while ignoring electrical noise on the I2C bus.

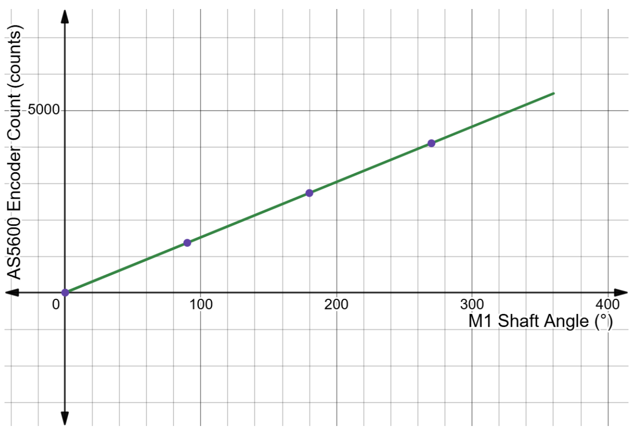


Figure 7: AS5600 raw count vs. M1 shaft angle (modulo 360°). With GR = 2, each 45° carousel step corresponds to 90° of motor shaft rotation and 1024 encoder counts. The index points are marked in the period.

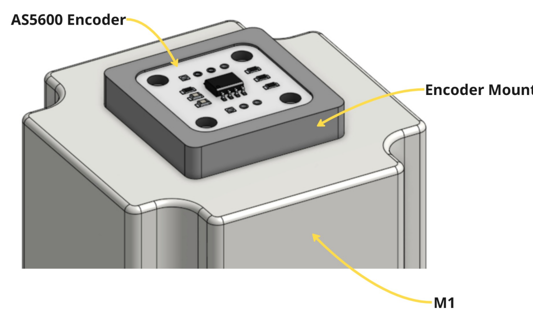


Figure 8: AS5600 encoder module mounted below M1, with the diametrically magnetised disc magnet press-fit to the motor shaft. A PLA magnet holder maintains the required 0.5 mm axial air gap.

On power-up, the firmware executes a homing routine: M1 rotates slowly until U6 reports the absolute angle corresponding to Module 1's pre-defined shaft reference. Because the AS5600 is an absolute encoder it reports position immediately on power-up without needing to traverse a limit switch or beam breaker. This sets the zero reference for all subsequent index moves. After every index move the firmware reads U6 and compares the measured shaft angle to the expected target. If the deviation exceeds ± 23 counts, corrective microsteps are issued until the position is within tolerance. A mandatory 1-second settling delay is then enforced before any dispense begins, allowing carousel vibration to fully damp and the load cell reading to stabilise.

Carousel Drive Schematic. The schematic below shows the complete electrical connections for Sub-Circuit 2: U1 signal pins to U4 driver inputs, U4 high-current outputs to M1 motor coils, and the I2C bus from U1 to the U6 encoder module.

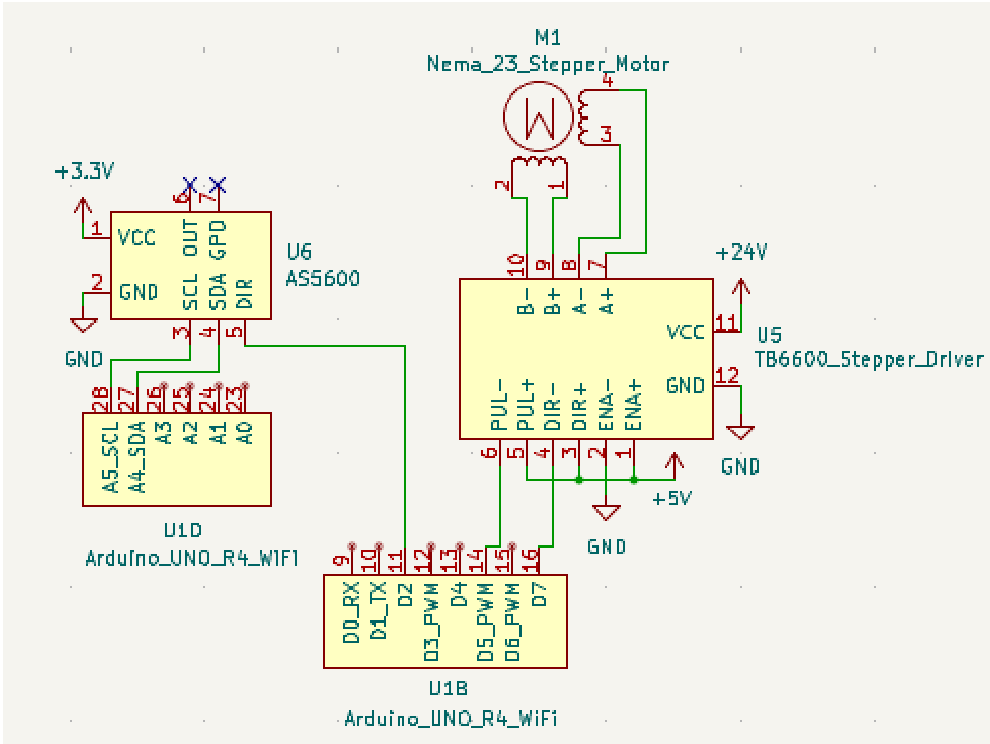


Figure 9: Carousel Drive schematic. U1 pins D3/D4 drive U4's PUL−/DIR− inputs via common-anode wiring. U4 supplies bipolar current to M1's coil pairs A and B. U6 is connected to U1's I2C bus (A4/A5) and powered from U1's 3.3 V regulator.

Firmware: Homing and Index Routines. The following two listings implement the carousel homing sequence (Listing 1) and the closed-loop index move with encoder verification (Listing 2). Both routines use the AccelStepper library for non-blocking step generation and the RobTillaart AS5600 library for I2C encoder reads.

```
void homeCarousel() {
  as5600.begin();
  carouselStepper.setMaxSpeed(200);
  carouselStepper.setSpeed(200);
  while (as5600.rawAngle() != MODULE_1_ANGLE) {
    carouselStepper.runSpeed();
  }
  carouselStepper.stop();
  carouselStepper.setCurrentPosition(0);
}
```

Listing 1: Carousel homing routine. M1 rotates slowly until U6 reports Module 1's known absolute shaft angle, then zeroes the step counter.

```
void indexCarousel(int targetSlot) {
  // GR = 2: each 45 deg carousel step = 1024 encoder counts
  long targetCounts = (long)(targetSlot - 1) * 1024L;
```

```
  carouselStepper.moveTo(targetCounts);
  while (carouselStepper.distanceToGo() != 0)
    carouselStepper.run();

  // Closed-loop correction: tolerance +/- 23 counts
  long measured = as5600.rawAngle();
  long error = targetCounts - measured;
  while (abs(error) > 23) {
    carouselStepper.move(error > 0 ? 1 : -1);
    carouselStepper.runSpeedToPosition();
    measured = as5600.rawAngle();
    error = targetCounts - measured;
  }
  delay(1000);  // 1-second settling delay before dispense
}
```

Listing 2: Carousel index move with closed-loop correction via U6. Corrective microsteps are issued if the measured angle deviates beyond $\pm$23 counts from the target. A 1-second settling delay follows every successful index.

2.2.3 Sub-Circuit 3: Auger Engagement Drive (M2, U5)

Once the carousel has indexed the correct container to the active position and the 1-second settling delay has elapsed, the auger engagement drive takes over. This sub-circuit meters spice from the indexed container by running a fixed NEMA 17 stepper motor (M2) that drives each container's internal auger screw via a half-spur-gear engagement mechanism.

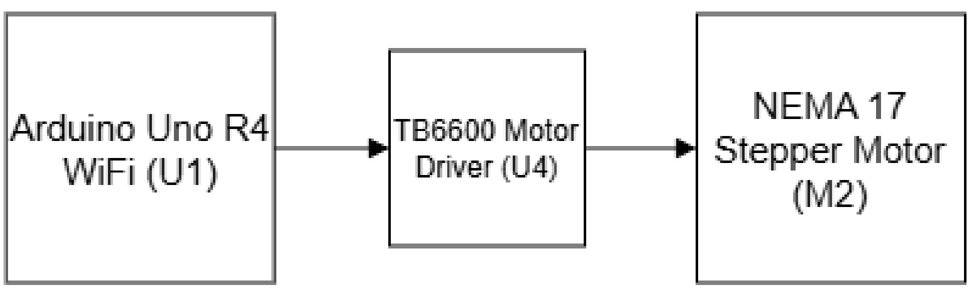


Figure 10: Auger Engagement Drive block extracted from the high-level block diagram. U1 pins D5 and D7 send STEP and DIR pulses to TB6600 driver U5, which drives NEMA 17 motor M2. M2 is fixed to the tower frame; it engages the active container's auger via a half spur gear that provides a positive mechanical cutoff when stopped.

M2 is mounted permanently to the tower frame below the carousel plane with its output shaft pointing vertically upward. M2 does not rotate with the carousel; it stays fixed and engages whichever container the carousel has brought to the dispensing position. The signal path has the same structure as the carousel drive (U1 $\rightarrow$ U5 $\rightarrow$ M2), but M2 operates entirely open-loop with respect to shaft angle — the only feedback that matters is the weight reading from Sub-Circuit 4. M2 runs until the load cell reports the target gram amount has been reached, then stops.

Motor Selection. A stepper motor was chosen over a DC motor because a stepper provides an instantaneous, firm stop with no coasting when the pulse train ceases. In a gram-accurate dispensing context, coasting after the stop command is the primary source of overshoot — a stepper eliminates this entirely. NEMA 17 was chosen over NEMA 23 because the auger load is mechanically very light; the motor only needs to overcome the rolling friction of dry spice powder against the auger helix and tube wall, with no significant inertial load. The smaller NEMA 17 form factor also simplifies the fixed mount above the carousel.

Parameter	Value
Step angle	1.8° (200 full steps/rev)
Rated phase current	1.5 A
Holding torque	45 N·cm
Shaft profile	5 mm D-profile
Microstepping (U5)	1/8 step ⇒ 1600 microsteps/rev

To confirm that M2 is adequately sized, the torque required to drive the auger against the worst-case powder load is estimated. For a half-filled auger tube with inner radius $r = 10$ mm, powder bulk density $\rho = 500$ kg/m^3, PLA-to-powder coefficient of friction $\mu = 0.4$, and auger helix pitch $p = 10$ mm, the powder volume in contact with one helix pitch is:

$$V_{\text{tube}} = \pi r^2 p = \pi(0.010)^2(0.010) = 3.14 \times 10^{-6}\ \text{m}^3$$

The torque required to push this powder through the helix is approximately:

$$\begin{aligned} T_{\text{auger}} &\approx \frac{1}{2}\mu\rho V_{\text{tube}} g r \\ &= 0.5 \cdot 0.4 \cdot 500 \cdot 3.14 \times 10^{-6} \cdot 9.81 \cdot 0.010 \\ &= 3.08 \times 10^{-3}\ \text{N} \cdot \text{m} \\ &= 0.31\ \text{N} \cdot \text{cm} \end{aligned}$$

M2 provides 45 N·cm holding torque — a $145\times$ margin over the worst-case auger load. This ensures M2 will never stall even with compacted powder or a more tightly packed helix than assumed.

Half Spur Gear Engagement Mechanism. The original HRS (HRS-3.1.14, HRS-3.1.26, HRS-3.1.27) specified a spring-loaded jaw coupler on M2's shaft engaging a D-shaft or hex stub on the top of each container. This approach is functionally valid but introduces alignment sensitivity: small deviations in container height or shaft positioning due to FDM print tolerances can lead to incomplete coupling or wear over repeated cycling.

The refined design uses a half spur gear mounted on M2's shaft. A half spur gear has teeth machined on only 180° of its circumference; the remaining 180° is a smooth, toothless arc.

Each container has a full spur gear on its top face, connected via a horizontal hex shaft to the internal auger screw. As the carousel brings a container to the active position, its full spur gear naturally comes into contact with M2's half spur gear. When M2 rotates, the toothed half engages the container's full spur gear and transfers rotation through 90° to the hex shaft, which turns the auger. When M2 stops with the toothless half facing the container spur gear, torque transfer ceases completely — providing a positive mechanical cutoff that prevents passive dripping regardless of powder pressure inside the container. No spring compliance, vertical actuation, or precise axial engagement is needed; the engagement occurs automatically as a natural consequence of the carousel bringing the container spur gear into the fixed plane of M2's half spur gear.

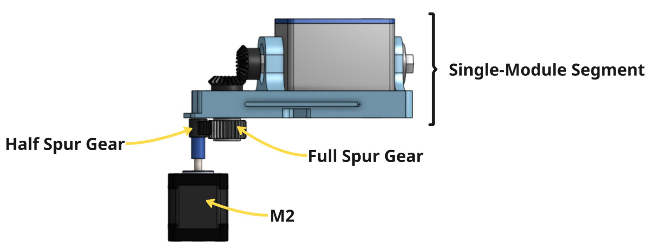


Figure 11: Side-view CAD render of the engagement mechanism. The half spur gear on M2's shaft (bottom) contacts the full spur gear on the container face (top). The toothed arc drives the auger; the toothless arc provides mechanical cutoff.

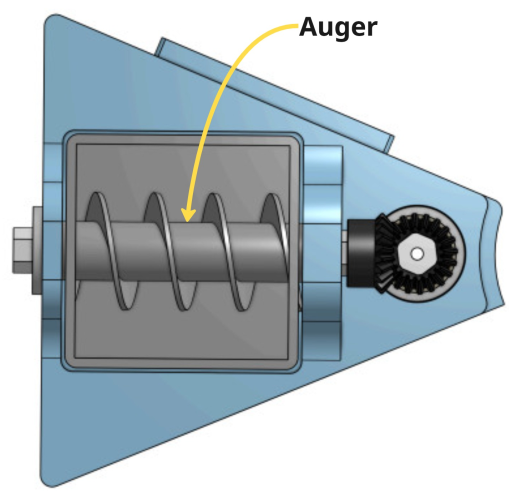


Figure 12: Top-view CAD render of the engagement mechanism, showing the auger screw inside the container body and the hex shaft connecting it to the spur gear face.

TB6600 Driver (U5) Wiring. U5 is wired identically to U4 in common-anode configuration, with STEP on U1 pin D5 and DIR on U1 pin D7. The current limit DIP switches on U5 are set to 1.5 A to match the rated phase current of the 17HS4401S, and microstepping is set to 1/8 step.

U5 Pin	Connection
ENA+	+5 V
ENA−	GND (always enabled)
PUL+	+5 V
PUL−	U1 pin D5
DIR+	+5 V
DIR−	U1 pin D7
VCC	+24 V rail
GND	System GND
A+, A−	M2 coil A
B+, B−	M2 coil B

Auger Engagement Drive Schematic. The schematic below shows the complete electrical connections for Sub-Circuit 3. The layout mirrors Sub-Circuit 2 with U5 replacing U4 and M2 replacing M1, reflecting the identical driver architecture used for both motors.

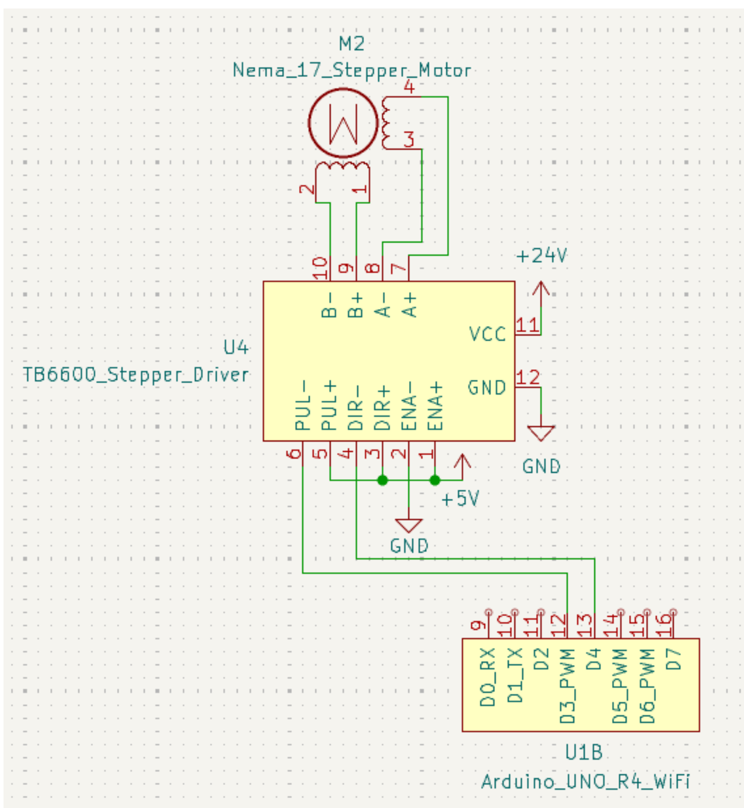


Figure 13: Auger Engagement Drive schematic. U1 pins D5/D7 drive U5's PUL−/DIR− inputs via common-anode wiring. U5 supplies bipolar current to M2's coil pairs. The ENA− pin is tied permanently to GND, keeping the motor energised and holding torque active at all times.

Three-Stage Speed Ramp-Down. Because in-flight spice remains inside the auger tube after M2 stops, running at full speed all the way to the target weight would cause consistent overshoot. The firmware implements a three-stage ramp-down keyed to the ratio of current dispensed weight to target weight:

Weight Threshold	Speed	Purpose
$W < 0.80\,W_{\text{target}}$	100%	Maximum throughput
$0.80 \leq W < 0.95\,W_{\text{target}}$	50%	Reduce in-flight spice volume
$0.95 \leq W < 1.00\,W_{\text{target}}$	15%	Fine approach, single-gram resolution
$W \geq W_{\text{target}}$	Stop	Coil current removed after 500 ms

Coil current is removed 500 ms after the stop command to prevent unnecessary heating of M2 during the idle period while the carousel indexes to the next container. The HX711 load cell reading is the sole criterion for stopping — not a step count or a timer. If the target weight is not reached within 60 s (indicating an empty container, a jammed auger, or a scale fault), the firmware halts M2, reports a timeout to Flask, and aborts the remainder of the blend.

```
void runDispense(float targetGrams) {
  unsigned long start = millis();
  while (true) {
    float w = scale.get_units(4);        // 4-sample average
    float ratio = w / targetGrams;
    if      (ratio < 0.80f) augerStepper.setMaxSpeed(FULL_SPEED);
    else if (ratio < 0.95f) augerStepper.setMaxSpeed(FULL_SPEED *
        0.50f);
    else if (ratio < 1.00f) augerStepper.setMaxSpeed(FULL_SPEED *
        0.15f);
    else {
      augerStepper.stop();
      delay(500);
      augerStepper.disableOutputs();
      return;
    }
    augerStepper.runSpeed();
    if (millis() - start > 60000UL) {
      augerStepper.stop();
      augerStepper.disableOutputs();
      sendJsonResponse("timeout", scale.get_units(4));
      abortBlend = true;
      return;
    }
  }
}
```

Listing 3: Three-stage speed ramp-down with 60-second per-spice timeout. The HX711 4-sample average is the sole stopping criterion.

2.2.4 Sub-Circuit 4: Weight Sensing (LC1, U3)

Every dispense in Bland2Grand is controlled by weight, not by time or step count. This is essential because the eight spices in the system have bulk densities ranging from 0.19 g/mL (oregano) to 0.55 g/mL (paprika) — a nearly 3:1 ratio. A volume-based or step-count-based approach would require a separate calibration curve for every spice at every fill level. A gram-accurate scale eliminates this variability entirely: Regardless of which spice is being dispensed or how densely it is packed, the system simply runs the auger until the scale reports the correct mass.

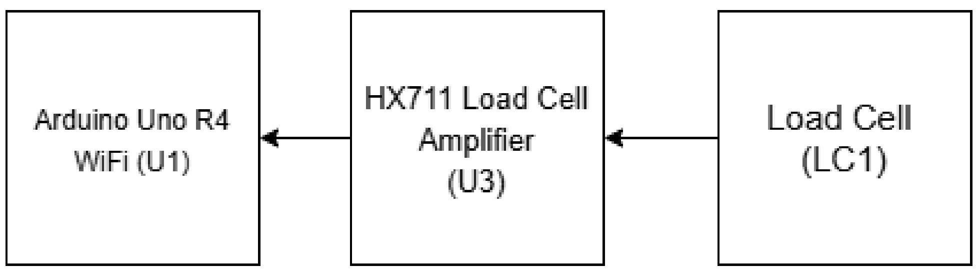


Figure 14: Weight Sensing block extracted from the high-level block diagram. The NOYITO load cell (LC1) is a full-bridge strain gauge whose differential millivolt output is amplified and digitised by the HX711 module (U3). U3 sends 24-bit weight data to U1 via a 2-wire bit-banged protocol on digital pins D9 (data) and D10 (clock).

The sensing chain works as follows: The bowl sits on the scale platform, supported by the NOYITO 1 kg full-bridge strain gauge load cell (LC1). As spice accumulates in the bowl, the flexure element inside LC1 bends under the increasing load. This bending strains four resistive gauges bonded to the flexure, changing their resistance by a tiny amount. The HX711 24-bit ADC module (U3) amplifies and digitises the resulting differential voltage and sends it to U1 over a 2-wire bit-banged serial interface on pins D9 and D10.

Wheatstone Bridge Theory. The four strain gauges inside LC1 are connected in a full Wheatstone bridge. Two gauges (R1, R4) are bonded in the tension direction — their resistance increases under load. Two gauges (R2, R3) are bonded in the compression direction — their resistance decreases under load. The bridge output voltage is:

$$V_{\text{out}} = V_{\text{ex}} \left(\frac{R_1}{R_1 + R_2} - \frac{R_3}{R_3 + R_4} \right)$$

At zero load, all four resistances are equal ($R_1 = R_2 = R_3 = R_4 = R$) and $V_{\text{out}} = 0$ (balanced bridge). Under load, R_1 and R_4 increase by ΔR while R_2 and R_3 decrease by

ΔR:

$$V_{\text{out}} = V_{\text{ex}} \cdot \frac{2\Delta R}{R + \Delta R} \approx V_{\text{ex}} \cdot \frac{2\Delta R}{R} \quad \text{(for small } \Delta R/R)$$

This is linear with applied force. The full-bridge configuration also provides inherent temperature compensation: all four gauges experience the same ambient temperature shift, so thermally-induced resistance changes cancel in the differential output. This is a critical property for a kitchen counter appliance that may operate across a range of room temperatures.

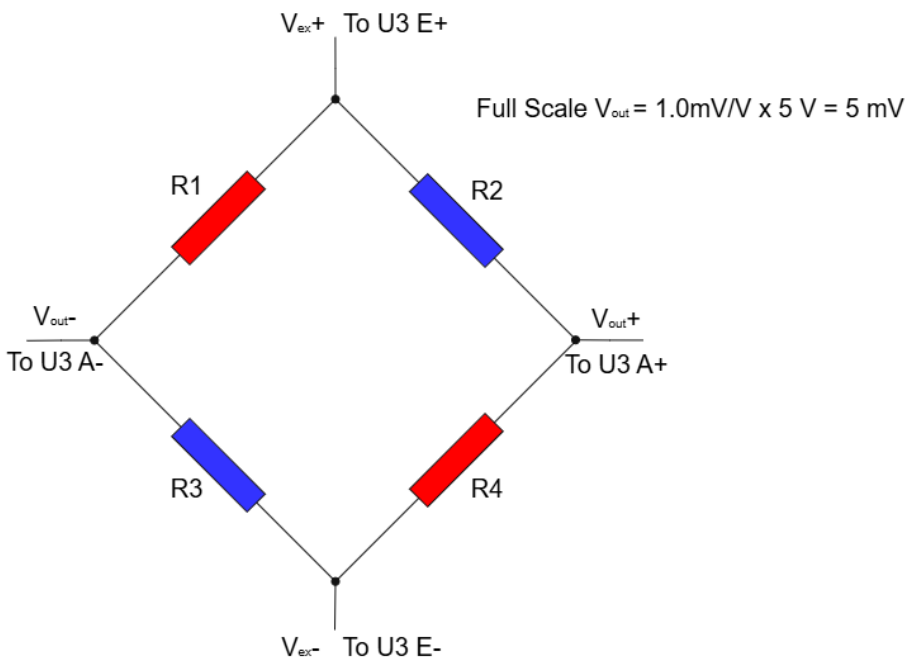


Figure 15: Full Wheatstone bridge inside LC1. R1 and R4 are in tension (resistance increases under load); R2 and R3 are in compression (resistance decreases). The differential output V_{out} is linear with applied force and self-compensating for ambient temperature changes.

Parameter	Value
Rated capacity	1000 g
Output sensitivity	1.0 mV/V
Excitation voltage	5 V (supplied by U3)
Full-scale output	1.0 mV/V $\times$ 5 V $=$ 5 mV
Safe overload	150% of rated (1500 g)
Wire colours	E$+$ red, E$-$ black, A$+$ green, A$-$ white
Max cable length to U3	200 mm

HX711 Amplifier and ADC. The full-scale bridge output is only 5 mV — far too small for

a standard ADC to resolve accurately. The HX711 (U3) is a dedicated load cell interface IC that solves this with an onboard instrumentation amplifier set to a fixed gain of 128$\times$ on Channel A, boosting the 5 mV full-scale signal to approximately 640 mV before the 24-bit sigma-delta ADC digitises it. The theoretical resolution is:

$$2^{24} = 16{,}777{,}216\,\text{counts} \qquad \Rightarrow \qquad \frac{1000\,\text{g}}{16{,}777{,}216} \approx 0.06\,\text{mg/count}$$

In practice, switching noise from the TB6600 drivers U4 and U5 couples into the load cell signal path and raises the effective noise floor to approximately 0.1 g per single sample. The cable between LC1 and U3 must not exceed 200 mm, as longer cables act as antennas for the 150 kHz switching frequency of U2 and the higher-frequency switching transients from U4 and U5. The firmware addresses the noise floor by averaging $N = 4$ consecutive readings before making any control decision:

$$\sigma_{\text{avg}} = \frac{\sigma_{\text{single}}}{\sqrt{N}} = \frac{0.1\,\text{g}}{\sqrt{4}} = 0.05\,\text{g}$$

This 0.05 g effective noise floor sits $6\times$ inside the ± 0.3 g system accuracy specification, providing comfortable margin. The RATE pin of U3 is tied to GND, selecting 10 SPS mode; 80 SPS mode triples the noise floor and offers no benefit at the slow dispense rates used here.

Platform Tilt and Cosine Error. The scale platform must remain level to within 2°. A tilted platform means the load cell measures only the component of gravitational force perpendicular to the sensing axis, introducing a cosine error:

$$\varepsilon = 1 - \cos(2) = 1 - 0.99939 = 0.061\%$$

On a 100 g measurement this produces a $\Delta W = 0.061$ g error — well within the ± 0.3 g specification. At 5° tilt the error grows to 0.38%, approaching the accuracy limit, so the flatness of the tower base is a meaningful mechanical requirement.

Two-Point Calibration. The HX711 outputs raw ADC counts proportional to load. To convert counts to grams, a two-point calibration is performed once per container slot: The scale is tared (zeroed) with an empty bowl, then a known standard weight is placed on the platform and the ADC count is recorded. The calibration factor is:

$$\text{cal\_factor} = \frac{\text{ADC}_{\text{loaded}} - \text{ADC}_{\text{tare}}}{\text{known\_weight (g)}} \qquad \Rightarrow \qquad \text{grams} = \frac{\text{ADC} - \text{ADC}_{\text{tare}}}{\text{cal\_factor}}$$

The calibration factor is stored in the Flask SQLite database and applied to all subsequent dispenses from that slot. Because each spice has a different bulk density and therefore different auger behaviour, a fresh tare is performed before every individual spice dispense — after the carousel has finished indexing and the 1-second settling delay has elapsed — to zero out any residual drift and ensure only the weight added by the current spice is measured.

HX711 Pin Connections.

Pin	Name	Connection
1	VSUP	+5 V rail
2	BASE	GND
3	AVDD	+5 V rail
5	AGND	GND
7	INA−	LC1 A− (white wire)
8	INA+	LC1 A+ (green wire)
11	PD_SCK	U1 D10
12	DOUT	U1 D9
15	RATE	GND (selects 10 SPS)
16	DVDD	+5 V rail

Calibration Curve and Load Cell Schematic. Figure 16 shows the expected linear relationship between weight and ADC count, with the ± 0.3 g accuracy limits marked as dashed bounds. Figure 17 shows the complete Weight Sensing schematic, and Figure 18 shows the load cell CAD mounting arrangement.

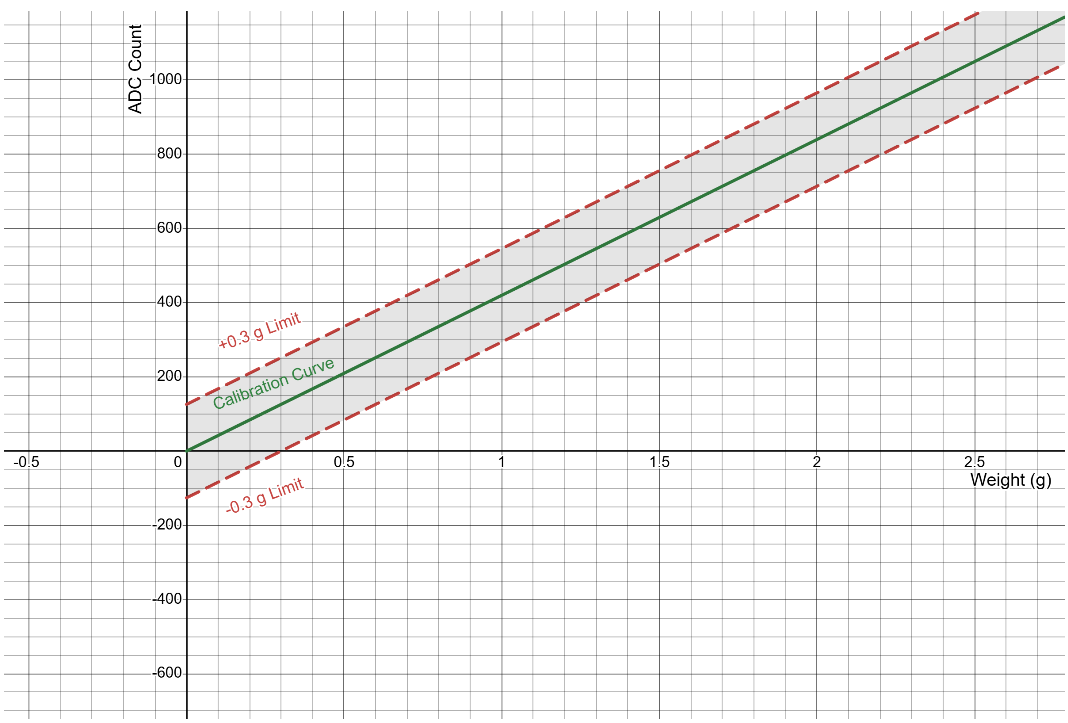


Figure 16: Linear calibration relationship between weight and ADC count. Dashed lines represent the ± 0.3 g system accuracy limits; the shaded region indicates the acceptable measurement band. All readings within this band meet the design specification.

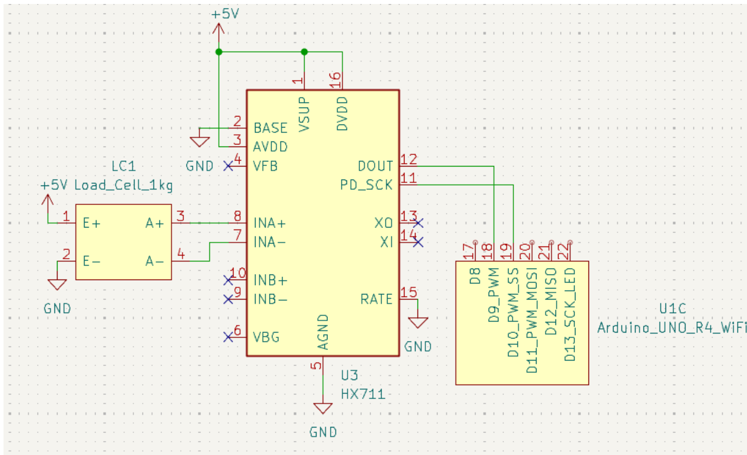


Figure 17: Weight Sensing schematic. LC1's four-wire bridge connects to U3's excitation and signal inputs. U3's DOUT and PD_SCK pins connect to U1 D9 and D10 respectively. The RATE pin is tied to GND to select 10 SPS mode.

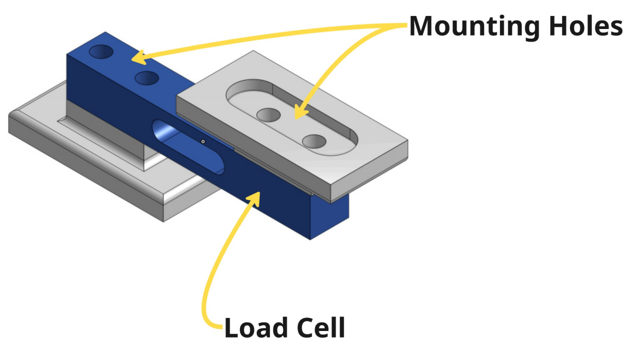


Figure 18: Isometric CAD view of the load cell mounting assembly. The NOYITO 1 kg load cell (LC1) is secured via M4 bolts through the mounting holes on each end. The scale platform rests on the load cell's upper flexure surface.

Firmware: Calibration and Weight Read. Listing 4 implements the two-point calibration routine that stores a per-slot calibration factor to the Flask database. Listing 5 implements the tare-and-read sequence called before every individual spice dispense.

```
void calibrate(int slot, float knownGrams) {
  long tare = scale.read_average(10);
  Serial.println("Place known weight...");
  delay(5000);
  long loaded = scale.read_average(10);
  float factor = (float)(loaded - tare) / knownGrams;
  saveCalibrationFactor(slot, factor);
  Serial.print("Cal factor: "); Serial.println(factor);
}
```

Listing 4: Two-point calibration routine. The scale is tared empty, a known weight is applied after a 5-second prompt delay, and the resulting calibration factor is saved to the Flask SQLite database.

```
void tareScale() {
  scale.tare();
  delay(500);          // Allow scale to settle after tare
}

float getWeight() {
  return scale.get_units(4);   // Average of 4 readings
}
```

Listing 5: Tare before each spice dispense and 4-sample averaged weight read. The 500 ms delay after tare allows the scale to fully settle before dispensing begins.

2.2.5 Sub-Circuit 5: Software and Communication (U1, Flask)

The software sub-circuit is what transforms the collection of motors and sensors described in Sub-Circuits 1–4 into a user-facing appliance. It spans two physical platforms: the Arduino Uno R4 WiFi (U1) running embedded C++ firmware, and a host laptop running a Python 3 Flask web server. The two platforms communicate over the local 2.4 GHz 802.11 b/g/n WiFi network via HTTP with JSON payloads. The end user interacts exclusively through a standard mobile web browser — no app installation is required.

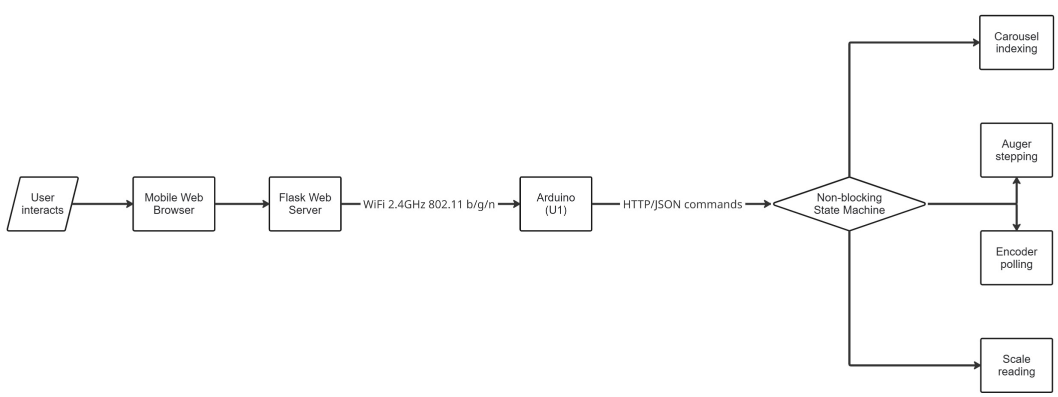


Figure 19: Software and Communication block and firmware state machine diagram. The user's phone communicates with the Flask server over WiFi. Flask orchestrates per-spice dispense commands to U1 over HTTP/JSON. U1 runs a non-blocking state machine that coordinates carousel indexing, auger stepping, encoder polling, and scale reading simultaneously without any blocking calls.

Arduino Uno R4 WiFi (U1). U1 is the central coordinator of the entire physical system. Its Renesas RA4M1 microcontroller (Arm Cortex-M4, 48 MHz, 32 KB SRAM, 256 KB Flash) runs the firmware that orchestrates all motor motion, reads all sensors, and handles WiFi communication. The hardware floating-point unit (FPU) on the Cortex-M4 accelerates the gram-target comparison arithmetic in the dispense loop without software overhead. The onboard ESP-S3 module handles 802.11 b/g/n WiFi on 2.4 GHz, accessed through the official WiFiS3 library. U1's full pin allocation across all sub-circuits is:

Pin	Net Label	Function
D3 (PWM)	STEP_M1	Carousel STEP pulse $\rightarrow$ U4 PUL$-$
D4	DIR_M1	Carousel DIR $\rightarrow$ U4 DIR$-$
D5 (PWM)	STEP_M2	Engagement STEP $\rightarrow$ U5 PUL$-$
D7	DIR_M2	Engagement DIR $\rightarrow$ U5 DIR$-$
D9	HX711_DAT	Data from U3 DOUT
D10	HX711_CLK	Clock to U3 PD_SCK
A4 (SDA)	I2C_SDA	U6 SDA (U6 module has onboard pull-ups)
A5 (SCL)	I2C_SCL	U6 SCL (U6 module has onboard pull-ups)
5V	PWR_5V	Input from U2 LM2596 output
3V3	PWR_3V3	Output to U6 VCC
GND	GND	Common system ground

Non-Blocking Firmware Architecture. The most important architectural constraint of the firmware is that it must *never block*. If the main loop stalls waiting for a motor to finish moving, the load cell stops being polled and the dispense overshoots. If it stalls waiting for a WiFi packet, the motor misses steps. The solution is a fully non-blocking event loop: `delay()` is never called in the main loop. All timing uses `millis()` comparisons. The AccelStepper library's `run()` method is called every loop iteration and generates step pulses non-blocking via software timing. The HX711 is polled every 100 ms; U6 encoder is polled every 50 ms; the WiFi client is checked every loop iteration. This ensures all sensing continues uninterrupted during motor operation, which is the fundamental requirement for gram-accurate closed-loop dispensing.

The firmware uses five open-source libraries, each chosen for confirmed compatibility with the RA4M1 platform:

Library	Purpose
AccelStepper	Non-blocking acceleration profiles for M1 and M2
RobTillaart AS5600	I2C encoder readout; confirmed RA4M1-compatible
bogde HX711	Load cell ADC reading; confirmed Uno R4 WiFi-compatible
WiFiS3	Official onboard WiFi library for Uno R4 WiFi
ArduinoJson	JSON serialisation and deserialisation

Firmware State Machine. The firmware operates as an explicit four-state machine driven entirely by sensor readings and timer comparisons — never by blocking calls:

- **IDLE** — waiting for an incoming WiFi command from Flask.
- **INDEXING** — carousel moving to the target slot with encoder verification.
- **DISPENSING** — auger running with scale feedback and three-stage ramp-down.
- **DONE** — sending the JSON response to Flask and taring the scale for the next spice.

A 30-second watchdog timeout returns the machine to IDLE if no command arrives from Flask during an active session, halting both motors safely.

Carousel Sequencing Algorithm. For a multi-spice blend, the firmware sorts the required slots by ascending carousel angle before issuing any motion commands, then rotates in one direction only. For example, a blend requiring slots [1, 4, 6, 8] is visited in that order ($1 \rightarrow 4 \rightarrow 6 \rightarrow 8$) rather than a naive order that could require reversals and extra travel. This minimises total carousel rotation time and ensures the blend completes well within the 120 s budget.

JSON Command Protocol. Flask and the Arduino communicate using a minimal JSON protocol over HTTP. Flask sends one command per spice, and the Arduino responds only after the dispense is complete (or timed out), so Flask's per-spice HTTP request naturally serialises the dispense sequence without any additional synchronisation logic.

Direction	Payload
Flask → Arduino	`{"carousel": N, "grams": X.X}`
Arduino → Flask (success)	`{"status": "done", "actual": Y.Y}`
Arduino → Flask (fault)	`{"status": "timeout", "actual": Y.Y}`

The `carousel` field specifies which of the 8 slots to index to; `grams` is the target mass for that spice. The `actual` field in the response reports the weight the scale measured at the moment M2 was stopped.

Flask Backend. The Flask server runs on Python 3 on the host laptop and is accessible from any device on the same WiFi subnet at `http://{laptop_ip}:5000`. A local SQLite database stores a minimum of 100 pre-built recipes. The five HTTP endpoints are:

Endpoint	Function
`GET /search`	Case-insensitive partial match; returns up to 3 blend options
`POST /dispense`	Accepts blend ID and serving count; orchestrates per-spice Arduino commands
`GET /status`	Returns current dispense progress as JSON (SSE or 2 Hz polling)
`POST /calibrate`	Accepts slot index and calibration factor; updates SQLite
`POST /recipe`	Saves user-defined custom recipe to SQLite

For dishes not found in the local SQLite database, Flask sends a structured prompt to the Anthropic API requesting a JSON spice blend for the named dish. The result is validated against the 8 available slots and saved to SQLite before being returned to the user. This save-on-first-use behaviour means the AI API is called at most once per unique dish name; all subsequent requests for the same dish are served locally in under 100 ms.

Firmware: Non-Blocking Main Loop. Listing 6 shows the main firmware loop. Every subsystem is polled or stepped on every iteration with no blocking calls.

```
void loop() {
  carouselStepper.run();
  augerStepper.run();

  if (millis() - lastScaleRead > 100) {
    currentWeight = scale.get_units(4);
    lastScaleRead = millis();
  }

  if (millis() - lastEncoderRead > 50) {
    encoderAngle = as5600.rawAngle();
    lastEncoderRead = millis();
  }

  WiFiClient client = server.available();
  if (client) handleClient(client);
```

```

  switch (state) {
    case IDLE:        handleIdle();       break;
    case INDEXING:    handleIndex();      break;
    case DISPENSING: handleDispense();  break;
    case DONE:        handleDone();       break;
  }
}
```

Listing 6: Non-blocking main loop. CarouselStepper and augerStepper are stepped every iteration via AccelStepper's non-blocking `run()` call. Scale and encoder are polled on separate timers. The state machine dispatcher executes the active state handler each iteration.

Flask: Dispense Orchestration and AI Fallback. Listing 7 shows the `/dispense` endpoint, which iterates over the sorted spice slots and issues one HTTP command to the Arduino per spice. Listing 8 shows the AI API fallback that generates a blend for any dish not already in the SQLite database.

```
@app.route('/dispense', methods=['POST'])
def dispense():
    data = request.get_json()
    blend = db.get_blend(data['blend_id'])
    servings = data['serving_count']
    for slot, grams_per_serving in sorted(blend.items()):
        target = round(grams_per_serving * servings, 1)
        if target == 0:
            continue
        resp = requests.post(ARDUINO_URL,
                             json={"carousel": slot, "grams": target},
                             timeout=90)
        result = resp.json()
        emit_sse(slot, result['actual'], target)
        if result['status'] == 'timeout':
            return jsonify({"error": f"Slot {slot} timed out"}), 500
    return jsonify({"status": "complete"})
```

Listing 7: Flask `/dispense` endpoint. Slots are sorted by ascending index, matching the carousel sequencing algorithm in the firmware. A per-slot 90-second HTTP timeout is set to match the 60-second firmware timeout with margin.

```
def get_recipe(dish_name):
    result = db.search(dish_name)
    if result:
        return result

```

```
    prompt = (f'Return ONLY a JSON object for a spice blend for '
              f'"{dish_name}" using slots: '
              f'1=Cumin, 2=Paprika, 3=GarlicPowder, 4=ChiliPowder, '
              f'5=Oregano, 6=OnionPowder, 7=BlackPepper, 8=Cayenne. '
              f'Format: {{"1": grams, "2": grams, ...}} for one serving.')
    response = anthropic.messages.create(
        model="claude-sonnet-4-6",
        max_tokens=200,
        messages=[{"role": "user", "content": prompt}]
    )
    blend = json.loads(response.content[0].text)
    db.save(dish_name, blend)
    return blend
```

Listing 8: AI API fallback with save-on-first-use. The Anthropic API is called at most once per unique dish name; the returned blend is saved to SQLite so all subsequent requests for the same dish are served locally.

2.3 Mechanical Assembly

The mechanical assembly provides the physical structure within which all electronic sub-circuits operate. It defines how M1's rotation reaches the carousel platform, how M2's fixed position above the carousel engages each container in turn, how dispensed spice reaches the bowl, and how the scale platform constrains the bowl in a level, repeatable position. Every dimension and tolerance in this section was chosen to support the functional requirements established in Sub-Circuits 1–5.

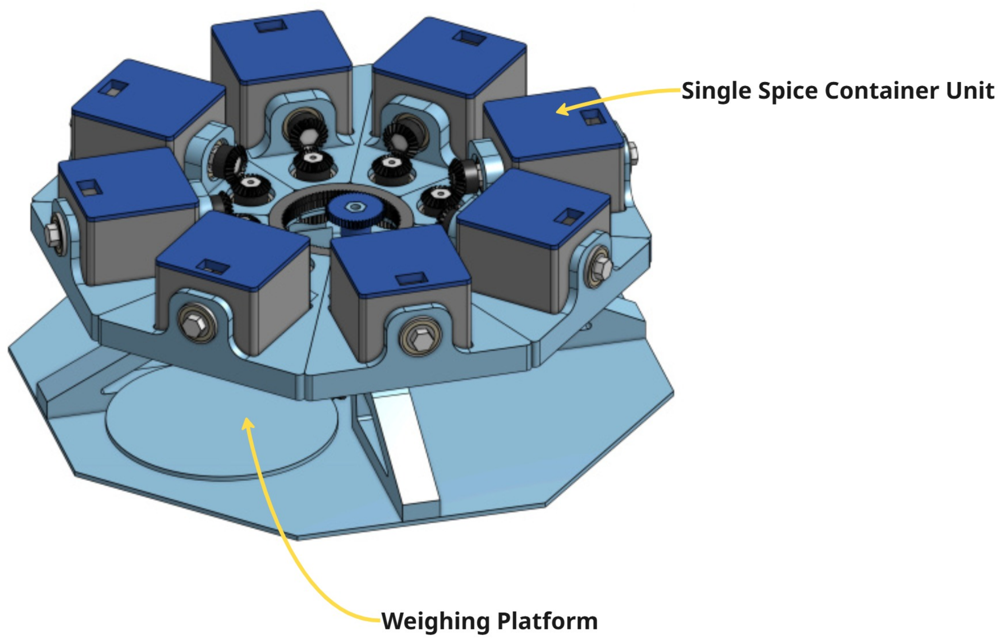


Figure 20: Isometric CAD overview of the complete Bland2Grand mechanical assembly, showing the eight-container carousel at mid-height, the output chute below the carousel plane, the scale platform at the base, and the tower enclosure that houses all electronics.

Tower Enclosure. The tower enclosure fits within the maximum envelope of 300 mm (W) $\times$ 300 mm (D) $\times$ 400 mm (H) specified in HRS-3.1.8. This envelope accommodates the octagonal carousel footprint, the motor mounts above and below it, the output chute, and the scale platform as an external-facing surface at the base. A USB Type-C port opening on the rear of the tower provides access to U1's programming interface. All structural and cosmetic components are printed in a single consistent PLA filament colour on the FRC team's 3D printers using the designer's personal filament.

Carousel Platform and Gear Train. The carousel platform is an octagonal PLA disc whose outer rim forms the ring gear that meshes with the pinion gear on M1's 8 mm shaft. As established in Sub-Circuit 2, the gear ratio is GR = 2: the ring gear has twice as many teeth as the pinion. In external mesh, the carousel rotates opposite to M1. This gear ratio provides mechanical advantage, the carousel receives twice the torque of M1 at half the speed, which is the correct trade-off for a heavy platform that must index smoothly without step loss.

Each of the 8 octagon faces features a quarter-turn bayonet locking recess for container mounting. Bayonet mechanisms are preferred over snap-fits here because bayonets distribute rotational force across the full circumference rather than concentrating it at a single snap point, and they do not fatigue under repeated cycling. The AS5600 encoder disc magnet is held by a small PLA press-fit magnet holder affixed to the underside of

M1's shaft end, maintaining the required 0.5 mm axial gap between the magnet and the U6 sensor PCB on the stationary motor bracket.

Spice Containers and Auger Mechanism. Each container is a square enclosure approximately 53.98 mm (2.125 in) tall. Off-the-shelf containers of equivalent geometry sourced from Walmart are used for the prototype; CAD dimensions confirm compatibility with the carousel bay geometry. Each container is fitted with a full spur gear on its top face for M2 engagement, a horizontal hex shaft running through the container interior, an auger screw on the hex shaft, and a downward-facing exit nozzle that aligns with the central tower chute when the container is in the active position.

The auger screw is printed with 0.2 mm radial clearance between the helix outer diameter and the tube inner wall — the minimum clearance that prevents binding from print layer irregularities, while being tight enough to prevent passive powder leakage when the auger is stationary. A circular flange disc at the motor-end of the helix section fits the tube with 0.1 mm radial clearance, acting as a physical seal that prevents fine powder from migrating up the shaft toward the spur gear engagement interface. All surfaces in the direct spice flow path are printed at a minimum of 50% infill density, ensuring non-porous walls that prevent hygroscopic spices from absorbing ambient moisture through the PLA matrix.

The auger tube is oriented at 20° above horizontal from the input end to the nozzle end. This upward angle means spice must travel uphill to exit the nozzle; gravity acts as a passive anti-drip valve whenever the auger is not turning. Combined with the half spur gear's toothless-arc cutoff described in Sub-Circuit 3, this provides two independent anti-drip mechanisms.

Output Chute. The fixed central output chute is integral to the tower frame and does not move. Its walls are angled at a minimum of 45° from vertical — 60–70° is preferred for fine powders such as garlic powder (bulk density 0.49 g/mL) which tend to bridge across shallower angles. The chute directs all dispensed spice to the centre of the LC1 scale platform below, ensuring the weight reading accurately captures every gram dispensed regardless of which container is active.

Scale Platform and Lid Retention. The LC1 scale platform is an external-facing horizontal surface at the tower base, sized to accommodate bowls up to 150 mm base diameter. It must remain level to within 2° to keep cosine error within the ± 0.3 g accuracy budget (calculated in Sub-Circuit 4). The platform flatness is constrained by the tower base geometry; no adjustment mechanism is required if the base is printed flat and the surface on which the tower rests is level.

The friction-fit lid on each container must resist opening during carousel rotation. At the maximum indexing speed of 60 RPM with containers at a radius of $r = 100$ mm, the centripetal acceleration on the lid is:

$$\omega = \frac{2\pi \times 60}{60} = 2\pi \text{ rad/s} \qquad \Rightarrow \qquad a = \omega^2 r = (2\pi)^2 \times 0.100 = 3.95 \text{ m/s}^2 \approx 0.40\,g$$

The lid must therefore resist a radial force of 0.40 times its own weight. A standard PLA friction-fit interference of 0.1–0.2 mm provides several newtons of retention force — more than sufficient at this acceleration level.

2.4 Product Interfaces

2.4.1 Internal Interfaces

The following tables document every signal connection internal to the Bland2Grand system. Each table corresponds to one logical signal path identified in the block diagram (Figure 1).

2.4.1.1 — Carousel Motor Interface (U1 → U4 → M1)

From	To	Signal Name	Description
U1-D3	U4-PUL−	STEP_M1	Step pulse, active-low
U1-D4	U4-DIR−	DIR_M1	Direction, active-low
+5V	U4-PUL+	—	Common anode reference
+5V	U4-DIR+	—	Common anode reference
+5V	U4-ENA+	—	Always enabled
GND	U4-ENA−	—	Enable reference
+24V	U4-VCC	—	Motor power
U4-A+	M1-A+	CAR_A+	Coil A positive
U4-A−	M1-A−	CAR_A−	Coil A negative
U4-B+	M1-B+	CAR_B+	Coil B positive
U4-B−	M1-B−	CAR_B−	Coil B negative

2.4.1.2 — Engagement Motor Interface (U1 → U5 → M2)

From	To	Signal Name	Description
U1-D5	U5-PUL−	STEP_M2	Step pulse, active-low
U1-D7	U5-DIR−	DIR_M2	Direction, active-low
+5V	U5-PUL+	—	Common anode reference
+5V	U5-DIR+	—	Common anode reference
+5V	U5-ENA+	—	Always enabled
GND	U5-ENA−	—	Enable reference
+24V	U5-VCC	—	Motor power
U5-A+	M2-A+	ENG_A+	Coil A positive
U5-A−	M2-A−	ENG_A−	Coil A negative
U5-B+	M2-B+	ENG_B+	Coil B positive
U5-B−	M2-B−	ENG_B−	Coil B negative

2.4.1.3 — AS5600 Encoder Interface (U1 ↔ U6, I2C) The U6 AS5600 module has onboard 4.7 kΩ pull-up resistors on SDA and SCL. No external pull-up resistors are required.

From	To	Signal Name	Description
U1-A4	U6-SDA	I2C_SDA	I2C data, 400 kHz
U1-A5	U6-SCL	I2C_SCL	I2C clock, 400 kHz
U1-3V3	U6-VCC	ENC_3V3	3.3 V encoder supply
GND	U6-GND	GND	Common ground

2.4.1.4 — Load Cell / HX711 Interface (LC1 → U3 → U1)

From	To	Signal Name	Description
LC1-E+	U3-E+ pad	CELL_EXC+	Bridge excitation + (red)
LC1-E−	U3-E− pad	CELL_EXC−	Bridge excitation − (black)
LC1-A+	U3-INA+ (pin 8)	CELL_SIG+	Signal positive (green)
LC1-A−	U3-INA− (pin 7)	CELL_SIG−	Signal negative (white)
U3-DOUT (pin 12)	U1-D9	HX711_DAT	Serialised weight data
U1-D10	U3-PD_SCK (pin 11)	HX711_CLK	Clock from U1 to U3

2.4.1.5 — Power Distribution Interface

From	To	Signal Name	Description
BT1-centre	24 V rail	PWR_IN	24 V from PSU
24 V rail	U4-VCC	PWR_24V	Carousel driver supply
24 V rail	U5-VCC	PWR_24V	Engagement driver supply
24 V rail	U2-IN+	BUCK_IN	Buck converter input
U2-OUT+	U1-5V pin	PWR_5V	5 V to Arduino
U2-OUT+	U3-VSUP	PWR_5V	5 V to HX711
U1-3V3	U6-VCC	PWR_3V3	3.3 V to encoder

2.4.2 External Interfaces

2.4.2.1 — AC Mains Power Input (BT1) The system accepts AC mains power (100–240 V, 50/60 Hz) via BT1, a 24 V 6 A AC/DC switching supply with a centre-positive 5.5 $\times$ 2.1 mm barrel jack output. A standard barrel jack cable is used; no custom connector is required.

Contact	Signal	Description
Centre pin	$+24\,\mathrm{V}$ DC	Positive supply from AC/DC adapter
Outer sleeve	GND	Power return, common system ground
SW1 (inline)	PWR_SW	SPST rocker switch on $+24\,\mathrm{V}$ conductor; rated $\geq 5\,\mathrm{A}$ at $24\,\mathrm{V}$ DC

2.4.2.2 — USB Programming Port The Arduino Uno R4 WiFi has an onboard USB-C connector accessible through a rear panel opening in the tower enclosure. This port is used for firmware programming via the Arduino IDE and for serial debug output at 115200 baud. A standard USB-A to USB-C cable is used; no custom connector or cable is required.

Pin	Signal	Description
A1, B1	GND	Ground
A4, B4	VBUS	$+5$ V USB power
A6, B6	D$+$	USB 2.0 data positive
A7, B7	D$-$	USB 2.0 data negative
A5, B5	CC1, CC2	Configuration channel (orientation detect)
—	All others	Unused (USB 2.0 only)

2.4.2.3 — WiFi Network Interface U1 operates in WiFi station mode on the 2.4 GHz 802.11 b/g/n local network. U1 connects to the network using SSID and password compiled into the firmware and listens on port 80 for HTTP POST commands from Flask. The Flask server on the host laptop is accessible at `http://{laptop_ip}:5000` from any mobile browser on the same subnet.

Parameter	Value
Standard	IEEE 802.11 b/g/n
Frequency band	2.4 GHz
Mode	Station (client)
Protocol	HTTP over TCP/IP
Listening port	80 (Arduino), 5000 (Flask)
Payload format	JSON
Command	`{"carousel": N, "grams": X.X}`
Response (success)	`{"status": "done", "actual": Y.Y}`
Response (fault)	`{"status": "timeout", "actual": Y.Y}`

2.4.2.4 — Bowl / Scale Platform The LC1 scale platform is an external-facing horizontal surface at the tower base, sized to accommodate a bowl with a base diameter up to 150 mm. The platform must remain level to within 2°. At 2° tilt:

$$\varepsilon = 1 - \cos(2) = 0.00061 = 0.061\% \quad \Rightarrow \quad \Delta W = 0.061\,\text{g on a }100\,\text{g measurement}$$

This is within the ± 0.3 g specification.

3 Scheduling

3.1 Tasks

The project is divided into four sequential phases, each ending with a verifiable milestone. The phases are designed so that the electronics and software are validated independently before being integrated into the mechanical system, reducing late-stage debugging risk.

ID	Task	Description
PHASE 1 — Electronics Breadboarding		**April 1 – April 10, 2026**
1	Component Verification	Verify all parts against BOM before proceeding.
1a	Receive and check parts	Receive all ordered parts; check every item against parts list.
1b	Inspect and power-test	Inspect each component for visible damage; test-power Arduino Uno R4 WiFi via USB-C.

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ID	**Task**	**Description**
2	Power System Breadboard	Set LM2596 to exactly 5.0 V before connecting any load.
2a	Connect U2 input	Connect U2 (LM2596) input to 24 V bench supply; do not connect load yet.
2b	Adjust output voltage	Adjust trimmer pot until multimeter reads exactly 5.0 V on U2 output.
2c	Connect U1	Connect U2 5 V output to U1 5 V pin; verify LED powers on.
2d	Verify 3.3 V rail	Verify $+$3.3 V on U1 3V3 pin (for U6 supply); measure with multimeter.
3	Carousel Motor (M1 + U4)	Wire and verify NEMA 23 stepper with TB6600 driver.
3a	Wire U4 power and DIP	Wire U4 (TB6600) VCC to 24 V rail; set DIP switches: 1/8 microstep, 2.0 A limit.
3b	Wire signal inputs	Wire U4 signal inputs: PUL$+$ 5 V, PUL$-$ D3, DIR$+$ 5 V, DIR$-$ D4.
3c	Connect motor coils	Connect M1 (NEMA 23) coil wires to U4: A$+$, A$-$, B$+$, B$-$.
3d	Test sketch	Write AccelStepper test sketch; run M1 CW/CCW at 60 RPM; confirm no missed steps.
4	Encoder (AS5600)	Wire and verify magnetic encoder via I2C.
4a	Wire encoder	Connect U6: VCC to U1 3V3, SDA to A4, SCL to A5 (onboard pull-ups; no external resistors).
4b	Read angle	Write I2C sketch using RobTillaart AS5600 library; read raw angle over serial.
4c	Verify counts	Rotate shaft to 0°/45°/90°/180°/360°; verify counts match $(4096/360) \times$ angle ± 11.
5	Engagement Motor (M2 + U5)	Wire and test NEMA 17 stepper with TB6600 driver.
5a	Wire driver	Wire U5 (TB6600) VCC to 24 V; set DIP switches: 1/8 microstep, 1.5 A limit.
5b	Wire signals	Wire U5 signal inputs: PUL$+$ 5 V, PUL$-$ D5, DIR$+$ 5 V, DIR$-$ D7.
5c	Connect coils	Connect M2 (NEMA 17) coil wires to U5: A$+$, A$-$, B$+$, B$-$.

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ID	**Task**	**Description**
5d	Test motion	Write AccelStepper test sketch; verify M2 CW/CCW operation at varying speeds.
6	Load Cell + HX711	Wire and calibrate weight sensing system.
6a	Wire HX711	Wire U3 (HX711): VSUP to 5 V, AGND to GND, RATE pin to GND (selects 10 SPS).
6b	Connect load cell	Connect LC1 4-wire to U3: E$+$ red, E$-$ black, A$+$ green, A$-$ white; DOUT$\rightarrow$D9, SCK$\rightarrow$D10.
6c	Verify ADC	Write bogde HX711 sketch; verify non-zero raw ADC; verify reading changes under load.
6d	Calibration	Two-point calibration: tare with empty bowl $+$ 200 g known weight; record cal_factor.
6e	Accuracy test	5 tare-and-weigh cycles with known weight; confirm $\leq \pm 0.3$ g.
7	Full Breadboard Integration	Combine all sub-circuits on breadboard into non-blocking main loop.
7a	Merge code	Combine all sketches into single non-blocking main loop using `millis()` (no `delay()`).
7b	Test system	Run combined sketch; verify U6/U3 reads every 100 ms; verify both motors step simultaneously.
M1 — All electronics functional on breadboard		
PHASE 2 — Software		**April 13 – April 26, 2026**
8	Flask Project Setup	Create Python 3 Flask backend with SQLite database.
8a	Create DB	Create Python 3 Flask folder; set up SQLite schema: dishes, blends, and calibration tables.
8b	Add data	Populate SQLite with 100$+$ dish recipes: Mexican, Italian, Indian/South Asian, BBQ/-Cajun.
9	Flask API Endpoints	Implement all five REST endpoints.
9a	Search endpoint	Implement `GET /search`: SQLite LIKE query; return up to 3 results as JSON.
9b	Dispense endpoint	Implement `POST /dispense`: accept blend_id $+$ serving_count; issue commands to U1.

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ID	**Task**	**Description**
9c	Status endpoint	Implement `GET /status`: return active_slot, dispensed_g, target_g for SSE polling.
9d	Calibration endpoint	Implement `POST /calibrate`: accept slot + grams_per_rev; update calibration table in SQLite.
9e	Recipe endpoint	Implement `POST /recipe`: accept custom blend name + 8 gram values; save to SQLite.
9f	AI fallback	Implement Anthropic API save-on-first-use: JSON blend prompt, validate, cache in SQLite.
10	U1 WiFi Firmware	Implement Arduino Uno R4 WiFi HTTP server on port 80.
10a	Connect WiFi	Write WiFiS3 HTTP server on U1: connect to WiFi SSID/password; listen on port 80.
10b	JSON handling	Implement ArduinoJson parsing: deserialise {`carousel, grams`}; serialise {`status, actual`}.
10c	Serial test	Validate Flask↔U1 over USB serial first; Flask sends JSON, U1 echoes parsed values.
10d	WiFi test	Transition to WiFi: confirm U1 on subnet; Flask HTTP POST to U1 port 80; verify round-trip.
11	Web Interface	Build mobile-first web UI for end users.
11a	Search UI	Build search screen: single input with 300 ms debounce autocomplete via `GET /search`.
11b	Results UI	Build results screen: 3 blend cards with spice amounts per serving + Select button.
11c	Serving slider	Build serving size slider (1–20) with client-side gram recalculation (no server round-trip).
11d	Progress UI	Build progress screen: per-spice bars, carousel slot indicator, and checkmarks on completion.
11e	Builder	Build custom recipe builder: blend name + 8 gram fields + Save via `POST /recipe`.
12	End-to-End Software Test	Full stack integration test.

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ID	**Task**	**Description**
12a	Full flow	Full flow test: phone $\rightarrow$ search $\rightarrow$ blend $\rightarrow$ serving $\rightarrow$ Flask $\rightarrow$ U1 $\rightarrow$ motor $\rightarrow$ response.
12b	Live updates	Verify SSE/2 Hz polling updates progress screen in real time during simulated dispense.
M2 — Full software stack functional over WiFi end-to-end		
PHASE 3 — Mechanical		**April 27 – May 17, 2026**
13	CAD Preparation	Prepare design tools and printing resources.
13a	CAD training	Complete CAD software training to proficient standard.
13b	Book printer	Book FRC printer time with Ben; confirm personal PLA filament on hand.
14	Tower and Chassis CAD	Model tower enclosure, motor mounts, chute, and scale platform.
14a	Tower enclosure	Design tower enclosure: 300×300×400 mm, USB-C cutout, chute opening, load cell bay.
14b	Output chute	Design output chute: $\geq$45° walls; directs spice to scale platform centre.
14c	Scale platform	Design load cell mounting plate and scale platform (accommodates 150 mm bowl).
15	Carousel Drive CAD	Model octagonal carousel platform with ring gear and bayonet bays.
15a	Pinion gear	Design pinion gear for M1 shaft (8 mm bore; tooth count for gear ratio GR = 2).
15b	Ring gear	Design ring gear on carousel outer circumference; confirm external mesh with pinion.
15c	Carousel platform	Design octagonal carousel: 8 bayonet bays, M1 shaft bore, ring gear rim.
15d	Magnet holder	Design U6 magnet holder: press-fit to M1 shaft, 0.5 mm axial gap to encoder PCB.
15e	M1 motor mount	Design M1 motor mount bracket below carousel plane.
16	Engagement Drive CAD	Model half spur gear, full spur gear, and container spur gear interface.
16a	Half spur gear	Design half spur gear for M2 shaft (teeth on 180° only; correct tooth profile).

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ID	**Task**	**Description**
16b	Full spur gear	Design full spur gear for container top face; mates with half spur gear; connects to hex shaft.
16c	Hex shaft housing	Design hex shaft housing inside each container; confirm alignment with auger input end.
16d	M2 fixed bracket	Design M2 fixed bracket: mounts to tower frame above carousel, shaft pointing downward.
17	Auger and Container CAD	Model trapezoidal containers with hex shaft, auger screw, and exit nozzle.
17a	Auger screw	Design auger screw: 0.2 mm radial clearance; flange seal disc 0.1 mm radial clearance.
17b	Auger tube	Design auger tube: 20° above horizontal from input to nozzle; downward exit nozzle at active face.
17c	Container body	Design container outer body: trapezoidal geometry, bayonet tabs, friction-fit lid seat.
17d	Friction-fit lid	Design friction-fit lid: positive retention; resists 0.40 g radial force at 60 RPM carousel speed.
18	CAD Assembly Verification	Verify full CAD assembly for interference and tolerance compliance.
18a	Interference check	Assemble all parts in CAD; check interference; confirm tolerances; verify spur gear mesh.
18b	Export renders	Export all CAD renders required for PDD documentation.
19	Test Prints	Print and validate critical parts before production run.
19a	Carousel section	Test print carousel section with one bayonet bay; verify bayonet tolerance and ring gear mesh.
19b	Gear pair	Test print half spur + full spur gear pair; verify mesh and torque transfer.

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ID	**Task**	**Description**
19c	Auger in tube	Test print auger screw in test tube; verify 0.2 mm clearance (spins freely, no passive leak).
19d	Revise and reprint	Revise CAD based on test print results; reprint any failed parts.
20	Production Prints and Assembly	Print all production parts and assemble tower.
20a	Carousel platform	Production print: full octagonal carousel platform.
20b	Gears	Production print: pinion gear + ring gear.
20c	Motor mounts	Production print: M1 motor mount + M2 fixed bracket.
20d	Spur gears	Production print: half spur gear ×2 + 8× full spur gear (one per container).
20e	Chute and platform	Production print: output chute + load cell platform.
20f	Magnet holder	Production print: U6 magnet holder.
20g	Source containers	Source 8 off-the-shelf containers from Walmart; verify geometry fits carousel bays.
20h	Prepare containers	Prepare containers: drill spur gear hole and auger exit nozzle in each container face.
20i	Carousel sub-assembly	Mechanical sub-assembly: carousel + M1 + pinion/ring gear + U6 magnet holder.
20j	Engagement sub-assembly	Mechanical sub-assembly: M2 fixed bracket + half spur gear.
20k	Install containers	Install 8 containers on carousel; confirm bayonet lock for all 8.
M3 — Mechanical system complete and assembled		
PHASE 4 — Integration		**May 18 – May 31, 2026**
21	Electronics Installation	Mount and wire all components inside tower.
21a	Mount electronics	Mount all electronics inside tower: U1, U4, U5, U3, U2.
21b	Route wiring	Route and dress all wiring: 18 AWG power, 22 AWG signal; cable-tie and label harnesses.
21c	Power-on safety check	Verify +24 V, +5 V, +3.3 V rails with multimeter before connecting motors.

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ID	**Task**	**Description**
22	Carousel and Encoder Validation	Verify all 8 index positions using AS5600 encoder feedback.
22a	Homing routine	Run carousel homing routine on full assembled tower; confirm Module 1 zero reference.
22b	Index verification	Step carousel through all 8 index positions; verify U6 reads correct count ($\pm$11).
23	Engagement Drive Validation	Confirm half spur gear engages cleanly with each container.
23a	Gear engagement	Index each container; confirm half spur engages full spur correctly for all 8 slots.
23b	Auger validation	Run M2 with spice; confirm auger feeds through chute; confirm stop halts spice immediately.
24	Per-Spice Calibration	Perform two-point calibration for all 8 spice slots.
24a	Load spices	Load all 8 containers with their respective spices.
24b	Record calibration	Run dispense calibration for each spice; record grams per 100 revolution-equivalents.
24c	Store calibration	Enter calibration data via `POST /calibrate`; update SQLite; verify stored values.
25	Closed-Loop Tuning	Optimise three-stage ramp-down thresholds for $\pm$0.3 g accuracy.
25a	Full end-to-end run	Full end-to-end: phone $\rightarrow$ WiFi $\rightarrow$ Flask $\rightarrow$ U1 $\rightarrow$ carousel $\rightarrow$ auger $\rightarrow$ bowl.
25b	Ramp-down tuning	Tune three-stage ramp-down thresholds (80%/50%, 95%/15%) per spice channel.
25c	Accuracy verification	10 consecutive dispenses per channel; confirm $\leq \pm 0.3$ g for all 8 channels.
26	Final System Verification	Verify full system performance against all HRS requirements.
26a	Timing	Run full 8-spice single-serving blend; confirm total dispense time $\leq$120 s.
26b	Safety	Simulate WiFi dropout (30 s watchdog); simulate empty container (60 s timeout).
26c	Access	Confirm USB-C port accessible from tower exterior for programming and serial debug.
M4 — Full integrated system verified, all HRS requirements confirmed		

3.2 Gantt Chart Schedule

Figure 21 shows the complete project schedule plotted as a Gantt chart. The four phases and their milestones are visible as colour-coded bars. Phase 1 and Phase 2 run sequentially with no overlap; Phase 3 begins immediately after M1 and runs in parallel with Phase 2's final integration testing. Phase 4 begins after M3 and concludes with M4 before the submission deadline.

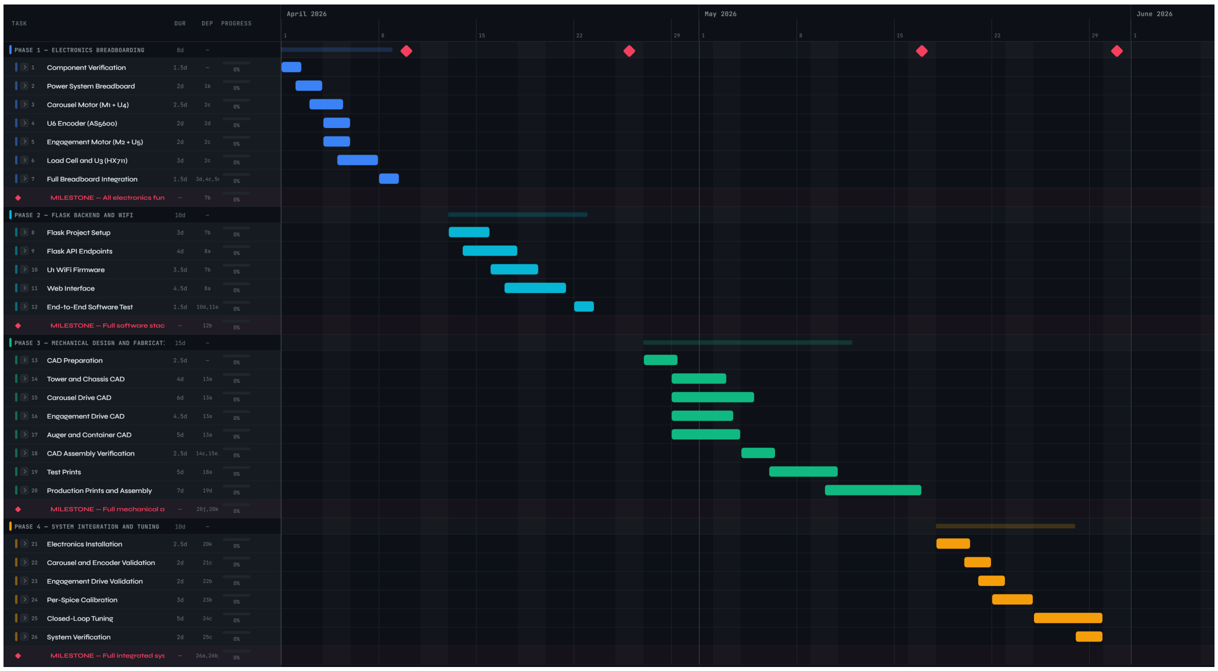

Figure 21: Bland2Grand project Gantt chart. Four phases are shown across April–May 2026. Milestones M1 (electronics breadboard complete), M2 (software functional), M3 (mechanical assembly complete), and M4 (full system verified) are marked with diamond symbols.

A Full System Schematic

The full system schematic consolidates all five sub-circuits onto a single sheet. It should be read in conjunction with the sub-circuit schematics presented inline in Section 2 (Figures 9, 13, and 17), which provide more legible detail for each individual signal path.

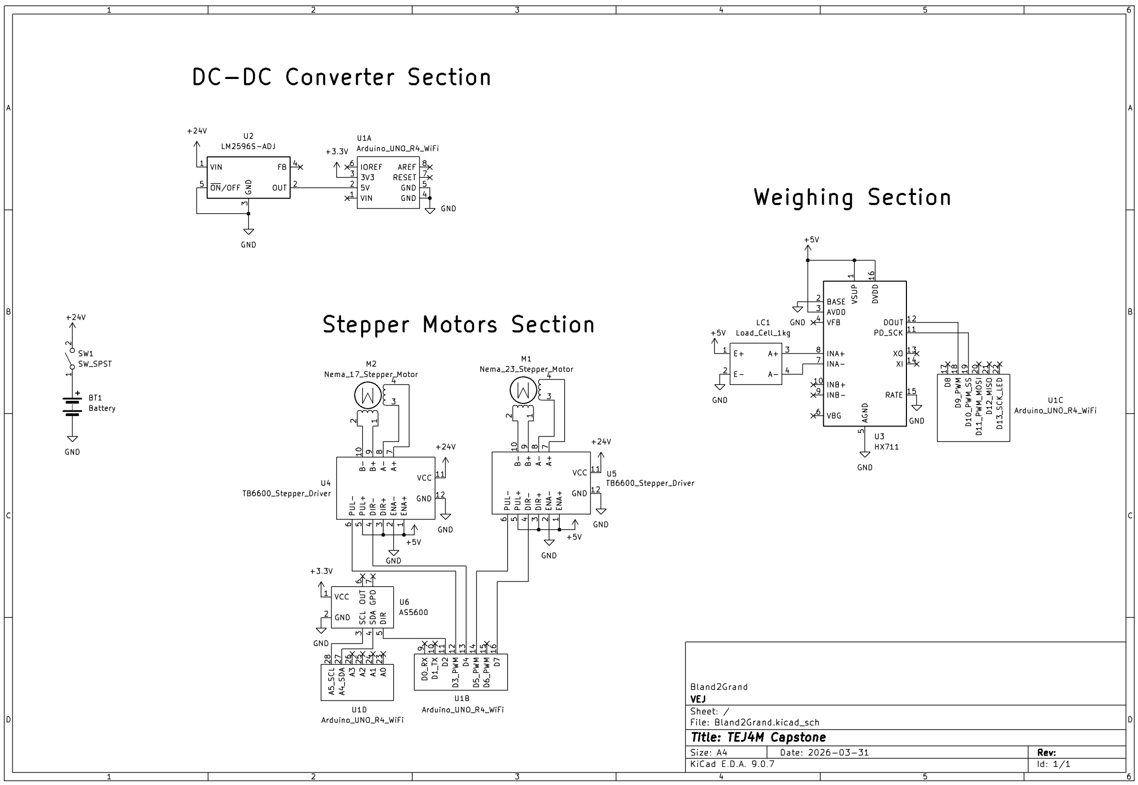

Figure 22: Overall Bland2Grand system schematic showing integration of all five sub-circuits.

B Parts List

Ref.	Qty	Part Description	Subsystem	Part / Kit No.	Owned	Link
U1	1	Arduino Uno R4 WiFi	Computing	ABX00087 (Uno R4 WiFi Kit)	Yes	Uno Kit
U2	1	MECCANIXITY LM2596S-ADJ DC-DC Buck Converter (2-pack)	Power	ASIN: B0DXZYL5CK	Yes	Link

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Ref.	**Qty**	**Part Description**	**Subsystem**	**Part / Kit No.**	**Owned**	**Link**
BT1	1	24V 6A AC/DC Power Supply with 5.5×2.1 mm Barrel Jack	Power	ASIN: B0BPLWKKWQ	Yes	Link
LC1+U3	1	NOYITO 1 kg Load Cell + HX711 24-bit ADC Module Combo	Weight Sensing	ASIN: B07BGRDKG5	Yes	Link
M1+U4	1	NEMA 23 23HS5628 Stepper Motor + TB6600 Driver Kit	Carousel Drive	ASIN: B0GD1GMX9T	No	Link
U6	1	AS5600 12-bit Absolute Magnetic Encoder Module with Disc Magnet (2-pack)	Carousel Drive	ASIN: B0DD7B64FB	Yes	Link
M2	1	NEMA 17 17HS4401S Stepper Motor	Auger Engagement	17HS4401S	No	Link
U5	1	TB6600 Stepper Motor Driver	Auger Engagement	ASIN: B0CXMC876T	Yes	Link
SW1	1	SPST Rocker Switch, $\geq 5\,\mathrm{A}$ at $24\,\mathrm{V}$ DC rated, panel-mount	Power	ASIN: B0FM7XSSN2	No	Link
—	1	PLA Filament 1.75 mm (approx. 500 g)	Mechanical	TBD	No	Brand TBD
—	8	Off-the-shelf spice containers (approx. 100 mm tall trapezoidal)	Mechanical	TBD	No	Walmart (TBD)
—	1	22 AWG Solid-Core Jumper Wire Assortment Kit	Wiring	Uno R4 WiFi Kit	Yes	Uno Kit
—	1	18 AWG Stranded Wire, Red and Black (power + motor leads)	Wiring	ASIN: B0746HMTPP	No	Link
—	1	USB-A to USB-C Cable (Arduino programming)	Computing	Uno R4 WiFi Kit	Yes	Uno Kit
—	1	5.5×2.1 mm Barrel Jack Cable (PSU to internal bus)	Power	Included w/ BT1	Yes	Included with PSU
—	4	M4 Bolts and Nuts assortment (load cell mounting)	Mechanical	N/A	No	Canadian Tire
—	1	Small Disc Magnet 6 mm diametrically magnetised	Carousel Drive	Included w/ U6	Yes	Included in U6 pack

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Ref.	**Qty**	**Part Description**	**Subsystem**	**Part / Kit No.**	**Owned**	**Link**
—	1	Multimeter (voltage verification during build)	Tools	N/A	Yes	—
—	1	Breadboard (Phase 1 electronics testing)	Testing	Uno R4 WiFi Kit	Yes	Uno Kit
—	1	Cable Ties / Zip Ties assortment (wiring harness management)	Wiring	N/A	Yes	—
—	1	Heat Shrink Tubing assortment (wire terminations)	Wiring	N/A	Yes	—
—	1	200 g Known Calibration Weight (HX711 two-point calibration)	Testing	N/A	Yes	—

C Definitions, Acronyms, and Abbreviations

Acronyms and Abbreviations

Acronym / Abbreviation	Definition
AC	Alternating Current — electrical current that periodically reverses direction; the form of power supplied from wall outlets.
ADC	Analog-to-Digital Converter — a circuit that converts continuous analog signals (such as voltage) into discrete digital numbers that a microcontroller can process.
AI	Artificial Intelligence — in this context, refers to the Anthropic Claude API used to generate spice blends for dishes not in the local database.
API	Application Programming Interface — a set of rules and protocols that allows one software application to communicate with another.
AS5600	The specific model of 12-bit absolute magnetic rotary encoder used for carousel position feedback.
AWG	American Wire Gauge — a standard system for specifying wire diameter; smaller numbers indicate thicker wire.
BOM	Bill of Materials — a comprehensive list of all components, parts, and materials required to build a system.
BT1	Reference designator for the main 24 V, 6 A AC/DC power supply.
CAD	Computer-Aided Design — software used to create 3D models and technical drawings of mechanical parts.

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Acronym / Abbreviation	**Definition**
CCW	Counter-Clockwise — direction of rotation opposite to the movement of clock hands.
CW	Clockwise — direction of rotation that follows the movement of clock hands.
DC	Direct Current — electrical current that flows in a single, constant direction.
DIR	Direction — a control signal that determines which way a stepper motor rotates.
DIP	Dual In-line Package — a type of switch housing with multiple small switches arranged in a row, used for configuration settings.
ENA	Enable — a control signal that activates or deactivates a motor driver's output.
FDM	Fused Deposition Modeling — the 3D printing technology used to fabricate plastic parts, including the carousel and gears.
Flask	A lightweight Python web framework used to build the backend server for Bland2Grand.
FPU	Floating-Point Unit — a hardware component within a microcontroller that accelerates mathematical operations with decimal numbers.
FRC	FIRST Robotics Competition — the high school robotics program whose 3D printers are used for this project.
GND	Ground — the common reference point for all voltages in an electrical circuit, typically at 0 V.
GR	Gear Ratio — the ratio of the number of teeth between two meshing gears; determines the relationship between input and output speed and torque.
HRS	Hardware Requirements Specification — the companion document that defines all *shall* requirements for the Bland2Grand system.
HTTP	Hypertext Transfer Protocol — the foundation of data communication on the World Wide Web; used for communication between Flask and the Arduino.
HX711	The specific model of 24-bit ADC module used to amplify and digitise the load cell signal.
I2C	Inter-Integrated Circuit — a two-wire serial communication bus used for connecting peripherals like the AS5600 encoder to the Arduino.
JSON	JavaScript Object Notation — a lightweight, human-readable data interchange format used for commands and responses between Flask and the Arduino.

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Acronym / Abbreviation	**Definition**
LC1	Reference designator for the NOYITO 1 kg load cell used for weight sensing.
LED	Light-Emitting Diode — a semiconductor device that emits light when current flows through it; used for status indication on the Arduino.
M1	Reference designator for the NEMA 23 stepper motor that drives the carousel rotation.
M2	Reference designator for the NEMA 17 stepper motor that drives the auger engagement mechanism.
MCU	Microcontroller Unit — a compact integrated circuit designed to control electronic systems; the RA4M1 is the MCU on the Arduino Uno R4 WiFi.
NEMA	National Electrical Manufacturers Association — sets standards for motor frame sizes; NEMA 17 and NEMA 23 refer to standardised mounting dimensions.
PCB	Printed Circuit Board — a mechanical assembly that electrically connects electronic components using conductive traces.
PDD	Preliminary Design Document — this document, which describes how the HRS requirements will be met.
PLA	Polylactic Acid — a biodegradable thermoplastic used for 3D printing the mechanical components.
PUL	Pulse — a control signal that causes a stepper motor to advance by one step or microstep.
PWM	Pulse Width Modulation — a technique for generating analog-like signals from digital outputs by rapidly switching a pin on and off.
RA4M1	The Renesas Arm Cortex-M4 microcontroller at the heart of the Arduino Uno R4 WiFi.
RPM	Revolutions Per Minute — a unit of rotational speed.
SCL	Serial Clock Line — the clock signal line in an I2C communication bus.
SDA	Serial Data Line — the data signal line in an I2C communication bus.
SPS	Samples Per Second — the rate at which an ADC converts analog signals to digital values.
SPST	Single-Pole Single-Throw — a simple on/off switch that connects or disconnects a single circuit.
SQLite	A lightweight, file-based relational database management system used to store recipes and calibration data.

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Acronym / Abbreviation	**Definition**
SSE	Server-Sent Events — a technology that allows a web server to push real-time updates to a client browser.
SW1	Reference designator for the main power rocker switch.
TB6600	The specific model of stepper motor driver used for both M1 and M2.
U1	Reference designator for the Arduino Uno R4 WiFi microcontroller board.
U2	Reference designator for the LM2596S-ADJ buck converter module that steps 24 V down to 5 V.
U3	Reference designator for the HX711 24-bit ADC module that interfaces with the load cell.
U4	Reference designator for the TB6600 driver controlling carousel motor M1.
U5	Reference designator for the TB6600 driver controlling auger motor M2.
U6	Reference designator for the AS5600 magnetic encoder module.
USB	Universal Serial Bus — a standard interface for connecting peripheral devices; the USB-C port on the Arduino is used for programming.
WiFi	Wireless Fidelity — a family of wireless networking protocols based on the IEEE 802.11 standard.

Technical Definitions

Term	Definition
Auger	A helical screw mechanism inside each spice container that rotates to push spice powder toward the exit nozzle; the speed and duration of rotation control the mass dispensed.
Bayonet Mount	A locking mechanism that uses a quarter-turn motion to secure the spice containers into the carousel; provides secure retention while allowing easy removal for refilling.
Buck Converter	A DC-DC step-down voltage regulator that efficiently converts a higher voltage (24 V) to a lower voltage (5 V) using switching technology rather than dissipating excess power as heat.
Carousel	The octagonal rotating platform that holds eight spice containers and indexes them to the dispensing position.

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Term	**Definition**
Centripetal Acceleration	The inward acceleration experienced by an object moving in a circular path; relevant for ensuring container lids remain secure during rotation.
Cosine Error	The measurement error introduced when a load cell platform is not perfectly level; the scale reads the component of gravitational force perpendicular to the sensing axis.
Full Bridge	A Wheatstone bridge configuration using four strain gauges that provides temperature compensation and twice the sensitivity of a half-bridge configuration.
Full Step	The basic step angle of a stepper motor when both phases are energised in the standard sequence; for the NEMA 23, one full step equals 1.8°.
Half Spur Gear	A gear with teeth machined on only 180° of its circumference; used on the auger motor shaft to provide a positive mechanical cutoff when the toothed arc is not engaged.
Holding Torque	The torque a stepper motor can maintain when energised but not rotating; essential for keeping the carousel stationary during dispensing.
Index	The act of rotating the carousel to bring a specific spice container into the active dispensing position.
Microstepping	A technique that divides each full stepper motor step into smaller increments by proportionally controlling the current in both motor phases; provides smoother motion and higher resolution.
NEMA 17	A standardised motor frame size with a 1.7-inch (43.2 mm) square mounting face; used for the auger drive motor M2 due to its compact size and adequate torque.
NEMA 23	A standardised motor frame size with a 2.3-inch (57.2 mm) square mounting face; used for the carousel drive motor M1 to handle the higher inertial load.
Pinion Gear	The small gear mounted on the M1 motor shaft that drives the larger ring gear on the carousel platform.
Ring Gear	The large gear that forms the outer circumference of the carousel platform; meshes with the pinion gear to provide mechanical advantage.

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Term	**Definition**
Settling Delay	A pause after carousel indexing that allows mechanical vibrations to damp and the load cell reading to stabilise before dispensing begins.
State Machine	A software architecture pattern where the system can be in one of a finite number of states (IDLE, INDEXING, DISPENSING, DONE) and transitions between them based on events.
Step Angle	The angular displacement of a stepper motor rotor for each full step; 1.8° for both M1 and M2, giving 200 full steps per revolution.
Strain Gauge	A sensor whose electrical resistance changes when mechanically deformed; four are arranged in a Wheatstone bridge inside the load cell to measure weight.
Wheatstone Bridge	An electrical circuit configuration used to measure small changes in resistance; the four strain gauges in the load cell form a full Wheatstone bridge.

Mechanical Fabrication Disclaimer

Mechanical Fabrication Disclaimer

The mechanical design for Bland2Grand was fully developed in CAD, including the carousel platform, pinion and ring gear assembly, trapezoidal spice containers, spur gear and hex shaft auger engagement mechanism, tower base, and output chute. Full-scale production 3D printing of all components in PLA with PCBWay would cost approximately $800 in filament and machine time, exceeding the $200 project budget.

Revised fabrication approach:

- **Spice containers:** Off-the-shelf containers of equivalent trapezoidal geometry sourced from Walmart or a similar retailer. CAD models confirm dimensional compatibility with the carousel bay geometry.
- **Tower base / enclosure:** A rigid structural enclosure (cardboard box or similar low-cost material) used as a functional prototype base. This does not affect any electronic or mechanical sub-circuit performance.
- **Printed parts:** The carousel platform, pinion and ring gear set, motor mounts, spur gear components, hex shaft housing, and output chute will be 3D printed on the FRC 3D printers using the designer's own filament, approved for use after consultation with Ms. O'Connor.

All mechanical dimensions referenced throughout this document and in the HRS remain valid. The fabrication method change does not alter any functional requirement, interface specification, or performance target.